Higher algebraic geometry

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1. Derived algebraic geometry

In this chapter, we follow the book [GR] of Gaitsgory-Rozenblyum, and give a brief account of what derived algebraic geometry is needed in geometric representation theory.

1.1. Stacks and schemes

1.1.1. Categorical constructions

Definition 1.1.1.1. Throughout this note, the word category will mean $(\infty, 1)$-categories unless stated otherwise. A functor is said to be continuous if it preserves all filtered colimits. Most importantly, we use cohomological indexing.

remark 1.1.1.1. We assume the reader is familiar with the definition of a t-structure on stable ∞-categories but slightly confused about the indexing convention, so we briefly recall here that what in the cohomological indexing convention should the t-structure should look like. For a stable ∞-category $\mathcal{C}$, it is equipped with two subcategories $\mathcal{C}^{\leqslant 0}$ and $\mathcal{C}^{\geqslant 0}$, where the shift or suspension functor $[1] = \Sigma$ preserves $\mathcal{C}^{\leqslant 0}$ (i.e., $[1]$ shifts the grading down). Moreover, we have $\tau^{\geqslant 0} : \mathcal{C} \to \mathcal{C}^{\geqslant 0}$ left adjoint to the inclusion and $\tau^{\leqslant 0} : \mathcal{C} \to \mathcal{C}^{\leqslant 0}$ right adjoint to the inclusion (which implies $\mathcal{C}^{\leqslant 0}$ is closed under all colimits and $\mathcal{C}^{\geqslant 0}$ is closed under all limits), which gives rise to the semi-orthogonality $H^0\mathrm{Hom}(c, d) = 0$ for $c \in \mathcal{C}^{\leqslant 0}$ and $d \in \mathcal{C}^{\geqslant 1}$. Then, we have the decomposition for any object

$$\tau^{\leqslant 0} c \to c \to \tau^{\geqslant 1} c$$

Finally, we may define the eventually connective or coconnective parts of $\mathcal{C}$ by

$$\mathcal{C}^- = \bigcup_{n \geqslant 0} \mathcal{C}^{\leqslant n}, \mathcal{C}^+ = \bigcup_{n \leqslant 0} \mathcal{C}^{\geqslant n}$$

We say a functor is left t-exact if it sends $\mathcal{C}^{\geqslant 0}$ to $\mathcal{D}^{\geqslant 0}$, and right t-exact if the dual version holds. Consequently, left t-exact functors sends $\mathcal{C}^+$ to $\mathcal{D}^+$ and right t-exact functors the dual.

Definition 1.1.1.2. We say a functor $F : \mathcal{C} \to \mathcal{D}$ is of finite tor-amplitude if it is left t-exact and right t-exact up to different shifts. We say an object $c \in \mathcal{C}$ a monoidal stable ∞-category is of finite tor-amplitude $- \otimes c : \mathcal{C} \to \mathcal{C}$ is so.

Definition 1.1.1.3. We fix a ground field k. We let $\mathbf{Vect} = \mathbf{Mod}_{Hk}(\mathbf{Sp})$ denote the category of module spectra of the Eilenberg-Maclane spectrum Hk regarded as a $\mathbb{E}_\infty$-ring spectrum.

Definition 1.1.1.4. We define the category $\mathbf{DGCat}$ to be $\mathbf{Mod}_{\mathbf{Vect}}(\mathbf{Pr}^L_{\mathrm{St}})$ (we will not go into details of the definitions and properties of $\mathbf{Pr}^L_{\mathrm{St}}$ here, but the readers may refer to [Lur08] for $\mathbf{Pr}^L$ and [Lur17] for $\mathbf{Pr}^L_{\mathrm{St}}$). Moreover, we can define its non-continuous version $\mathbf{DGCat}^{\mathrm{nc}}$, given as follows:

- We begin with the category $\mathbf{Mod}_{\mathbf{Vect}^\omega}(\mathbf{Cat})$.
- We take its full subcategory spanned by categories that are stable and having that the action functor $\mathbf{Vect}^\omega \times \mathcal{C} \to \mathcal{C}$ exact in both variables. We denote this category by $\mathbf{DGCat}^{\text{non-cocmpl}}$

- We define $\mathbf{DGCat}^{\mathrm{nc}}$ to be the essential image of the functor $\mathbf{DGCat} \to \mathbf{DGCat}^{\text{non-cocmpl}}$.

todo 1.1.1.1. Write something more here, at least explain left complete t-structure and blah blah.

1.1.2. The category of prestacks

Definition 1.1.2.1. We define the category of affine schemes by $\mathbf{Aff} = \mathbf{CAlg}(\mathbf{Vect}^{\leqslant 0})^{\mathrm{op}}$.

remark 1.1.2.1. The geometry of non-connective and connective schemes differ subtlely, and we (following Gaitsgory and possibly Lurie) consider the connective case in most circumstances.

Definition 1.1.2.2. We define the category of prestacks by $\mathbf{pStk} = \mathrm{pSh}(\mathbf{Aff}) = \mathrm{Fun}(\mathbf{Aff}^{\mathrm{op}}, \mathbf{Ani})$. By Yoneda embedding, we obtain an inclusion $\mathbf{Aff} \hookrightarrow \mathbf{pStk}$.

Definition 1.1.2.3. We now introduce a truncated version of affine schemes. Consider the subcategory of $\mathbf{Vect}$ given by the objects concentrated between degrees $[-n, 0]$, say $\mathbf{Vect}^{[-n,0]}$. We have a symmetric monoidal structure on this subcategory given by $(A, B) \mapsto \tau^{\geqslant -n}(A \otimes B)$, thus we may define the category of n-coconnective affine schemes ${}^{\leqslant n}\mathbf{Aff} = \mathbf{CAlg}\big(\mathbf{Vect}^{[-n,0]}\big)^{\mathrm{op}}$. This category of commutative algebras is denoted by $\mathbf{CAlg}^{[-n,0]}$ also for short.

Construction 1.1.2.4. Notice that the inclusion $\mathbf{Vect}^{[-n,0]} \hookrightarrow \mathbf{Vect}^{\leqslant 0}$ is lax symmetric monoidal, and inparticular preserves commutative algebras. Thus we have a fully faithful functor

$$\mathbf{CAlg}^{[-n,0]} \hookrightarrow \mathbf{CAlg}^{\leqslant 0}$$

hence an inclusion ${}^{\leqslant n}\mathbf{Aff} \hookrightarrow \mathbf{Aff}$.

Notice that we have a truncation functor $\tau^{\geqslant -n} : \mathbf{Vect}^{\leqslant 0} \to \mathbf{Vect}^{[-n,0]}$, which naturally restricts to the commutative algebra category, that is, we have a commutative diagram

$$\begin{array}{ccc} \mathbf{CAlg}^{\leqslant 0} & \xrightarrow{\tau^{\geqslant -n}} & \mathbf{CAlg}^{[-n,0]} \\ \Big\downarrow{\scriptstyle U} & & \Big\downarrow{\scriptstyle U} \\ \mathbf{Vect}^{\leqslant 0} & \xrightarrow[\tau^{\geqslant -n}]{} & \mathbf{Vect}^{[-n,0]} \end{array}$$

where U is the forgetful functor. Moreover, $\tau^{\geqslant -n}$ is left adjoint to the inclusion functor. We denote the functor induced on the category of affine schemes by taking op by ${}^{\leqslant n}\tau : \mathbf{Aff} \to {}^{\leqslant n}\mathbf{Aff}$. We abuse notations and write the same functor post-composed with the inclusion ${}^{\leqslant n}\mathbf{Aff} \hookrightarrow \mathbf{Aff}$.

Definition 1.1.2.5. We have ${}^{\leqslant 0}\mathbf{Aff}$ by definition the classical category of affine schemes. Hence, we will also denote ${}^{\leqslant 0}\tau(-)$ by $(-)_{\mathrm{cl}}$

Definition 1.1.2.6. We define the category

$$ {}^{<\infty}\mathbf{Aff} = \bigcup {}^{\leqslant n}\mathbf{Aff} $$

the category of eventually coconnective affine schemes.

Definition 1.1.2.7. We define the category of n-prestacks by ${}^{\leqslant n}\mathbf{pStk} = \mathrm{pSh}({}^{\leqslant n}\mathbf{Aff}^{\mathrm{op}})$. Restriction along the inclusion ${}^{\leqslant n}\mathbf{Aff} \hookrightarrow \mathbf{Aff}$ gives a functor ${}^{\leqslant n}\tau : \mathbf{pStk} \to {}^{\leqslant n}\mathbf{pStk}$. Moreover, the left Kan extension functor gives a fully faithful left adjoint of ${}^{\leqslant n}\tau$, say $\mathrm{LKE} : {}^{\leqslant n}\mathbf{pStk} \hookrightarrow \mathbf{pStk}$, which somehow justifies the name of the above truncation functor.

remark 1.1.2.2. Right Kan extension gives a right adjoint of the functor ${}^{\leqslant n}\tau$.

Warning 1.1.2.8. We should think ${}^{\leqslant n}\mathbf{pStk}$ as the prestacks n-coconnective in the sense that they are detected or determined by all the maps from n-coconnective affines, rather than the usual sense of truncatedness of presheaves, i.e., has values in n-coconnective vector spaces.

Definition 1.1.2.9. Let $\mathcal{X}$ be a prestack. We say $\mathcal{X}$ is convergent (or nilcomplete in the sense of [Lur18]) if for any $S \in \mathbf{Aff}$, we have $\mathcal{X}(S) \to \lim_n \mathcal{X}({}^{\leqslant n}\tau(S))$ is an equivalence.

Lemma 1.1.2.10. Any prestack representable by an affine scheme is convergent.

Proof: By completeness of the t-structure in $\mathbf{Sp}$ hence in $\mathbf{Vect}$, for any algebra $A \in \mathbf{CAlg}(\mathbf{Vect}^{\leqslant 0})$ we have $A \to \lim \tau^{\leqslant n} A$ is an isomorphism. The rest follows by Yoneda embedding and the universal property of $\tau^{\leqslant n}$ as a left adjoint. □

Example 1.1.2.3. The stack of isomorphisms of quasi-coherent sheaves is not convergent.

Proposition 1.1.2.11. A prestack $\mathcal{X}$ is convergent iff as a functor $\mathcal{X} : (\mathbf{Aff})^{\mathrm{op}} \to \mathbf{Ani}$, we have the equivalence $\mathcal{X} \simeq \mathrm{RKE}\big(\mathcal{X}|_{({}^{<\infty}\mathbf{Aff})^{\mathrm{op}}}\big)$

Proof: We first directly compute the right Kan extension of $\mathcal{Z} : ({}^{<\infty}\mathbf{Aff})^{\mathrm{op}} \to \mathbf{Ani}$. By the pointwise limit formula,

$$ \mathcal{Z}(S) = \lim_{S' \in ({}^{<\infty}\mathbf{Aff})_{/S}} \mathcal{Z}(S') $$

So it suffices to see that the functor $\mathbb{N} \to ({}^{<\infty}\mathbf{Aff})_{/S}$ by sending n to ${}^{\leqslant n}\tau(S)$ is cofinal in the category. We omit the higher bullshit here and verify only that it is cofinal on the level of underlying 1-categories. By definition, any object $T \in {}^{<\infty}\mathbf{Aff}$ is n-truncated for some n, and a map $T \to S$ hence factors through $T \to {}^{\leqslant n}\tau(S)$ by adjunction. □

Construction 1.1.2.12. By the above proposition, if we let $\mathbf{pStk}_{\mathrm{conv}} \subset \mathbf{pStk}$ denote the subcategory of convergent prestacks, this inclusion admit a left adjoint $\mathcal{X} \mapsto \mathcal{X}_{\mathrm{conv}}$, constructed as $\mathcal{X}_{\mathrm{conv}} = \mathrm{RKE}(\mathcal{X}|_{{}^{<\infty}\mathbf{Aff}})$

Beside coconnectivity, there is a parallel notion we can consider on stacks. We denote this property by truncatedness.

Definition 1.1.2.13. We say a prestack $\mathcal{X}$ is k-truncated if as a functor $\mathbf{Aff}^{\mathrm{op}} \to \mathbf{Ani}$, it has image in $\mathbf{Ani}_{\leqslant k}$. We denote this category by $\mathbf{pStk}_{\leqslant k}$. Likewise, we may define the category ${}^{\leqslant n}\mathbf{pStk}_{\leqslant k}$. This form a $(k,1)$-category.

Example 1.1.2.4. By definition, the Yoneda image of ${}^{\leqslant n}\mathbf{Aff}^{\mathrm{op}}$ in ${}^{\leqslant n}\mathbf{pStk}$ is n-truncated.

1.1.3. Finite type conditions

Definition 1.1.3.1. We say an object $A \in \mathbf{CAlg}^{\leqslant 0}$ is of finite type if $H^0(A)$ is of finite type over k and $H^i(A)$ is finitely generated module over $H^0(A)$. We denote this category by $\mathbf{CAlg}^{\leqslant 0}_{\mathrm{ft}}$. We denote the intersection $\mathbf{CAlg}^{\leqslant 0}_{\mathrm{ft}} \cap \mathbf{CAlg}^{[-n,0]} = \mathbf{CAlg}^{[-n,0]}_{\mathrm{ft}}$. Likewise on its opposite, we define ${}^{<\infty}\mathbf{Aff}_{\mathrm{ft}}$ and ${}^{<\infty}\mathbf{Aff}_{\mathrm{ft}} \cap {}^{\leqslant n}\mathbf{Aff} = {}^{\leqslant n}\mathbf{Aff}_{\mathrm{ft}}$.

Theorem 1.1.3.2 ([Lur17]). The objects of $\mathbf{CAlg}^{[-n,0]}_{\mathrm{ft}}$ are compact in $\mathbf{CAlg}^{[-n,0]}$. Moreover, for any $S \in {}^{\leqslant n}\mathbf{Aff}$, the category opposite to $I = ({}^{\leqslant n}\mathbf{Aff}_{\mathrm{ft}})_{/S}$ is filtered, and the cofiltered limit $S \mapsto \lim_{S_0 \in I} S_0$ is an isomorphism.

Corollary 1.1.3.3. $\mathbf{CAlg}^{[-n,0]}_{\mathrm{ft}}$ is a subcategory of compact generators in $\mathbf{CAlg}^{[-n,0]}$. Hence, $\mathbf{CAlg}^{[-n,0]} = \mathrm{Ind}\left(\mathbf{CAlg}^{[-n,0]}_{\mathrm{ft}}\right)$. Dually, we have ${}^{\leqslant n}\mathbf{Aff} = \mathrm{Pro}({}^{\leqslant n}\mathbf{Aff}_{\mathrm{ft}})$. Moreover, since $\mathbf{CAlg}^{[-n,0]}_{\mathrm{ft}}$ is idempotent complete, we have $\mathbf{CAlg}^{[-n,0]}_{\mathrm{ft}} = \left(\mathbf{CAlg}^{[-n,0]}\right)^{\omega}$.

Definition 1.1.3.4. Let $\mathcal{X} \in {}^{\leqslant n}\mathbf{pStk}$ for some n, we say it is locally of finite type if it is the left Kan extension $\mathcal{X} = \mathrm{LKE}\left(\mathcal{X}|_{{}^{\leqslant n}\mathbf{Aff}_{\mathrm{ft}}}\right)$. We denote the full subcategory by ${}^{\leqslant n}\mathbf{pStk}_{\mathrm{lft}}$.

Construction 1.1.3.5. Regarding the above definition as ${}^{\leqslant n}\mathbf{pStk}_{\mathrm{lft}} = \mathrm{pSh}({}^{\leqslant n}\mathbf{Aff}_{\mathrm{ft}})$, we may restate the above definition as there is an inclusion functor given by the left Kan extension ${}^{\leqslant n}\mathbf{pStk}_{\mathrm{lft}} \hookrightarrow {}^{\leqslant n}\mathbf{pStk}$ with a right adjoint given by restriction of sites.

Corollary 1.1.3.6. By Theorem 1.1.3.2, a stack $\mathcal{X} \in {}^{\leqslant n}\mathbf{pStk}$, equivalent to a functor $\mathbf{CAlg}^{[-n,0]} \to \mathbf{Ani}$ is left Kan extended from $\mathbf{CAlg}_{\mathrm{ft}}^{[-n,0]} = \left(\mathbf{CAlg}^{[-n,0]}\right)^{\omega}$ iff it preserves filtered colimits. Thus, by expanding definitions, ${}^{\leqslant n}\mathbf{Aff} \cap {}^{\leqslant n}\mathbf{pStk}_{\mathrm{lft}} = {}^{\leqslant n}\mathbf{Aff}_{\mathrm{ft}}$.

Definition 1.1.3.7. We say an affine scheme $S \in \mathbf{Aff}$ is almost of finite type if for all n we have ${}^{\leqslant n}\tau(S)$ is of finite type. We denote the full subcategory of almost of finite type affine schemes by ${}^{\leqslant n}\mathbf{Aff}_{\mathrm{aft}}$.

Dually, an algebra $A \in \mathbf{CAlg}^{\leqslant 0}$ is almost of finite type if $H^0(A)$ is of finite type over k and $H^n(A)$ is finitely generated as an $H^0(A)$-module (for example, this implies that A is coherent in the sense of [Lur17], Definition 7.2.4.16, since we are working over fields). We denote the category by $\mathbf{CAlg}_{\mathrm{aft}}^{\leqslant 0}$.

Definition 1.1.3.8. We say a prestack $\mathcal{X} \in \mathbf{pStk}$ is locally almost of finite type if it is convergent and for every n, we have ${}^{\leqslant n}\tau(\mathcal{X}) \in {}^{\leqslant n}\mathbf{pStk}_{\mathrm{lft}}$. We denote the corresponding full subcategory by $\mathbf{pStk}_{\mathrm{laft}}$.

Corollary 1.1.3.9. By Corollary 1.1.3.6, $\mathbf{pStk}_{\mathrm{laft}} \cap \mathbf{Aff} = \mathbf{Aff}_{\mathrm{aft}}$.

Proposition 1.1.3.10. We have an equivalence of categories

$$\mathbf{pStk}_{\mathrm{laft}} = \mathrm{pSh}({}^{<\infty}\mathbf{Aff}_{\mathrm{ft}})$$

where $\mathbf{pStk}_{\mathrm{laft}} \to \mathrm{pSh}({}^{<\infty}\mathbf{Aff}_{\mathrm{ft}})$ is given by restricting along the inclusion ${}^{<\infty}\mathbf{Aff}_{\mathrm{ft}} \hookrightarrow \mathbf{Aff}$, and the inverse is given by $\mathrm{RKE}_{({}^{<\infty}\mathbf{Aff} \hookrightarrow \mathbf{Aff})}\,\mathrm{LKE}_{({}^{<\infty}\mathbf{Aff}_{\mathrm{aft}} \hookrightarrow {}^{<\infty}\mathbf{Aff})} : \mathrm{pSh}({}^{<\infty}\mathbf{Aff}_{\mathrm{aft}}) \to \mathrm{pSh}({}^{<\infty}\mathbf{Aff}) \to \mathrm{pSh}(\mathbf{Aff})$.

Proof: Recall that we require a locally almost of finite type prestack $\mathcal{X}$ to be convergent. By Proposition 1.1.2.11, $\mathcal{X}$ is determined by its data on ${}^{<\infty}\mathbf{Aff}^{\mathrm{op}}$. It suffices to prove that the data of $\mathcal{X}$ on ${}^{<\infty}\mathbf{Aff}^{\mathrm{op}}$ is determined by its data on ${}^{<\infty}\mathbf{Aff}_{\mathrm{aft}}^{\mathrm{op}}$ via left Kan extension. By definition, the restriction ${}^{\leqslant n}\tau(\mathcal{X}) : {}^{\leqslant n}\mathbf{Aff}^{\mathrm{op}} \to \mathbf{Ani}$ is obtained from left Kan extension from ${}^{\leqslant n}\mathbf{Aff}_{\mathrm{ft}}^{\mathrm{op}}$. Thus, it suffices to identify the following two conditions on a convergent stack $\mathcal{X}$ regarded as a functor $\mathcal{X} : \mathbf{CAlg}^{\leqslant 0} \to \mathbf{Ani}$:

- It is a Kan extension along $\mathbf{CAlg}_{\mathrm{aft}}^{\leqslant 0} \hookrightarrow \mathbf{CAlg}^{\leqslant 0}$.
- Its restriction to any $\mathbf{CAlg}^{[-n,0]}$ is the Kan extension along $\mathbf{CAlg}_{\mathrm{ft}}^{[-n,0]} \hookrightarrow \mathbf{CAlg}^{[-n,0]}$.

Notice that by definition $\mathbf{CAlg}^{[-n,0]} \cap \mathbf{CAlg}_{\mathrm{aft}}^{\leqslant 0} = \mathbf{CAlg}_{\mathrm{ft}}^{[-n,0]}$, then note for $A \in \mathbf{CAlg}^{[-n,0]}$, $B \in \mathbf{CAlg}^{\leqslant 0}$, we can compute $H^0\mathrm{Hom}(B, A) = H^0\mathrm{Hom}(\tau^{\geqslant -n}B, A)$.

We first show the first condition implies the second. In the pointwise colimit formula for computing the value at A, namely

$$\mathrm{LKE}(F)(A) = \operatorname*{colim}_{B \to A} F(B)$$

we may assume $B \in \mathbf{CAlg}_{\mathrm{ft}}^{[-n,0]}$ since any map $B \to A$ factors through $B \to \tau^{\geqslant -n}B \to A$, exhibiting the subcategory $\left(\mathbf{CAlg}_{\mathrm{ft}}^{[-n,0]}\right)_{/A}$ cofinal in $\left(\mathbf{CAlg}_{\mathrm{aft}}^{\leqslant 0}\right)_{/A}$. Thus, the first condition clearly implies the second.

The converse is done similarly. □

The above arguments give a basic setting for performing descent to extend convergent functors from affine schemes to prestacks.

1.1.4. Descent conditions and stacks

Definition 1.1.4.1. Recall a morphism $A \to B$ of $\mathbb{E}_\infty$-rings in $\mathbf{Vect}^{\leqslant 0}$ is flat if $H^0(A) \to H^0(B)$ is flat, and the map $H^n(A) \otimes_{H^0(A)} H^0(B) \to H^n(B)$ is an isomorphism. For $S, S' \in \mathbf{Aff}$, we say a morphism $S \to S'$ is flat if it is so on the structure rings.

By definition, the notion of flatness is convergent, i.e., a map $S \to S'$ is flat iff ${}^{\leqslant n}\tau(S) \to {}^{\leqslant n}\tau(S')$ is so.

Definition 1.1.4.2. We say a morphism of affine schemes $f : S \to S'$ is ppf, smooth, étale, open embedding, Zariski if f is flat and f_{cl} is so.

We say furthermore f is a flat, smoooth, étale, Zariski covering if f_{cl} is moreover surjective.

Definition 1.1.4.3. We say a morphism of prestacks $f : \mathcal{X} \to \mathcal{Y}$ is affine schematic, if for any affine scheme $S \to \mathcal{Y}$, we have the pullback $S \times_{\mathcal{Y}} \mathcal{X}$ affine.

Definition 1.1.4.4. We say an affine schematic map of prestack $\mathcal{X} \to \mathcal{Y}$ is flat, ppf, smooth, étale, open, Zariski if the pullback map is so for any $S \to \mathcal{Y}$ where S is affine.

Definition 1.1.4.5. We say a prestack $\mathcal{X}$ satisfies flat, ppf, smooth, étale, Zariski descent if the following conditions holds.
(1) $\mathcal{X}(\varnothing) = *$.
(2) $\mathcal{X}(S_1 \sqcup S_2) = \mathcal{X}(S_1) \times \mathcal{X}(S_2)$.
(3) For $S' \to S$ a covering, the map $\mathcal{X}(S) \to \mathrm{Tot}(\mathcal{X}(S'^{\bullet}/S))$ is an isomorphism.

We define the category **Stk** (we omit the topology in the notation however) to be the full subcategory of **pStk** with objects satisfying the above descent conditions.

Definition 1.1.4.6. We fix a topology $T \in \{\text{Zariski, étale, smooth, ppf, flat, ...}\}$. We define a map $\mathcal{X} \to \mathcal{Y}$ to be a T-equivalence if for any $\mathcal{Z} \in \mathbf{Stk}$ with respect to T, we have an isomorphism $\mathrm{Hom}(\mathcal{Y}, \mathcal{Z}) \to \mathrm{Hom}(\mathcal{X}, \mathcal{Z})$.

Construction 1.1.4.7. The inclusion $\mathbf{Stk} \hookrightarrow \mathbf{pStk}$ admit a left adjoint, which is the localization functor at the T-equivalences. We denote this localization by $L : \mathbf{pStk} \to \mathbf{Stk}$, and we sometimes abuse notations and write $L : \mathbf{pStk} \to \mathbf{Stk} \to \mathbf{pStk}$. We call it the sheafification functor. By construction, it commutes with finite limits.

Lemma 1.1.4.8. We fix a topology $T \in \{\text{Zariski, étale, ppf, flat}\}$. Then, let $f : \mathcal{X} \to \mathcal{Y}$ be an T-surjection. Then, we have

$$|\mathrm{Bar}(\mathcal{X}^\bullet/\mathcal{Y})| \to \mathcal{Y}$$

is a T-equivalence.

Proof: By [Lur08] Corollary 6.2.3.5, it suffices to prove that $\tau_{\leqslant -1}(\mathcal{X} \to \mathcal{Y})$ is the final object of $\mathbf{pStk}_{/\mathcal{Y}}$. This is direct by surjectivity. □

Corollary 1.1.4.9. Let $\mathcal{X} \to \mathcal{Y}$ be a T-surjection. Then, we have an equivalence of categories

$$\big|\mathrm{Bar}(\mathbf{Stk}^\bullet_{\mathcal{X}}/\mathbf{Stk}_{\mathcal{Y}})\big| = \mathbf{Stk}_{\mathcal{Y}}$$

Proof: By Lemma 1.1.4.8, we have $\mathcal{X} \to \mathcal{Y}$ an effective epimorphism in **Stk**. This then follows from the property that **Stk**, as a localization of **pStk** by a topology, is a topos. □

Corollary 1.1.4.10.

Lemma 1.1.4.11. The image of the Yoneda functor $\mathbf{Aff} \hookrightarrow \mathbf{pStk}$ lies in **Stk**.

Proof: Since $T = \text{flat}$ is the finest topology, it suffices to show that the flat topology is subcanonical. This is done by reduction to the ordinary faithfully flat descent as follows:

Take a faithfully flat map $A \to B$ and a ring R. We wish to prove that

$$\mathrm{Hom}_{\mathbf{CAlg}^{\geqslant 0}}(R, A) = \mathrm{Tot}_{n \geqslant 1}\big(\mathrm{Hom}_{\mathbf{CAlg}^{\geqslant 0}}(R, B^{\otimes_A n})\big)$$

Note $\mathrm{Hom}_{\mathbf{CAlg}^{\geqslant 0}}(R, -)$ preserves limits, it suffices to prove

$$A = \lim_{\Delta}\big(B \rightrightarrows B \otimes_A B \;\substack{\rightarrow\\\rightarrow\\\rightarrow}\; \cdots\big)$$

We prove that this is an isomorphism on each cohomology. By flatness, we have

$$H^n(B^{\otimes_A m}) = H^n(A) \otimes_{H^0(A)} H^0(B)^{\otimes_{H^0(A)} m}$$

Then, since taking limits commutes with taking cohomology, we have the map

$$H^n(A) \to \lim_{\Delta} H^n(A) \otimes_{H^0(A)} H^0(B)^{\otimes_{H^0(A)} m} = H^n(A)$$

by the classical faithfully flat descent. Hence the Yoneda functor has image in **Stk**. □

Lemma 1.1.4.12. Consider the Cartesian square

$$\begin{array}{ccc} \mathcal{X}' & \longrightarrow & \mathcal{X} \\ \downarrow & \ulcorner & \downarrow \\ S' & \xrightarrow{f} & S \end{array}$$

in **Stk** such that $S, S', \mathcal{X}' \in \mathbf{Aff}$. Moreover, assume f is a T-covering. Then, $\mathcal{X} \in \mathbf{Aff}$.

Proof: Assume $S' \to S = \operatorname{Spec} B \to \operatorname{Spec} A$ induced by a faithfully flat map $f : A \to B$. By an analogous proof of Lemma 1.1.4.11, we have

$$\mathbf{Mod}_A = \lim\Big(\mathbf{Mod}_B \rightrightarrows \mathbf{Mod}_{B\otimes_A B} \substack{\rightarrow\\\rightarrow\\\rightarrow} \cdots\Big)$$

Passing to commutative algebras, we have

$$\mathbf{CAlg}_A = \lim\Big(\mathbf{CAlg}_B \rightrightarrows \mathbf{CAlg}_{B\otimes_A B} \substack{\rightarrow\\\rightarrow\\\rightarrow} \cdots\Big)$$

Then, assume $\mathcal{X}' = \operatorname{Spec} T$ for some $T \in \mathbf{CAlg}_B$. We define $R \in \mathbf{CAlg}_A$ given by the descent

$$R = \lim\Big(S \rightrightarrows S \otimes_B B \otimes_A \otimes B \substack{\rightarrow\\\rightarrow\\\rightarrow} \cdots\Big)$$

where the structure morphisms of the cosimplicial object is given by the pullback map above, essentially because $\mathcal{X}'$ is the pullback of something (not yet known to be affine) over S. We now prove $\mathcal{X} = \operatorname{Spec} R$. This is because $\mathcal{X}$ and $\operatorname{Spec} R$ share the same descent datum, and hence they are the same by Corollary 1.1.4.9. □

Corollary 1.1.4.13. Let $f : \mathcal{X} \to \mathcal{Y}$ be an affine schematic morphism in **pStk**. Then, the sheafification $L(f) : L(\mathcal{X}) \to L(\mathcal{Y})$ is an affine schematic morphism.

Proof: Take any $S \in \mathbf{Aff}$ and $S \to L(\mathcal{Y})$, and we wish to show the product $S \times_{L(\mathcal{Y})} L(\mathcal{X})$ is affine. By Lemma 1.1.4.12, it suffices to show that for a T-surjection $S' \to S$ we have the fiber product $S' \times_{L(\mathcal{Y})} L(\mathcal{X})$ is affine. Note the unit $\mathcal{Y} \to L(\mathcal{Y})$ is T-surjective, so we may choose S' such that the composition $S' \to S \to L(\mathcal{Y})$ factors through $\mathcal{Y}$ Since L commutes with finite limits and S' is affine hence a stack, it suffices to prove $S' \times_{L(\mathcal{Y})} L(\mathcal{X}) = L(S' \times_{\mathcal{Y}} \mathcal{X})$ is affine. However, $S' \times_{\mathcal{Y}} \mathcal{X}$ is affine, so applying L is an equivalence. □

The interaction of sheafification with n-coconnectivity is rather curious.

Definition 1.1.4.14. We denote ${}^{\leqslant n}\mathbf{Stk}$ the full subcategory of ${}^{\leqslant n}\mathbf{pStk}$ that satisfy T-descent in ${}^{\leqslant n}\mathbf{Aff}$. We denote ${}^{\leqslant n}L$ by the corresonding sheafification functor. By definition, the functor $\mathbf{pStk} \to {}^{\leqslant n}\mathbf{pStk}$ sends **Stk** to ${}^{\leqslant n}\mathbf{Stk}$, and preserves T-equivalences.

Construction 1.1.4.15 ([Lur17], 6.5.3). Recall that ${}^{\leqslant n}\mathbf{Stk}$ is the category of sheaves over an n-category ${}^{\leqslant n}\mathbf{Aff}$. Hence, the sheafification functor ${}^{\leqslant n}L$ is the composition of $(n+2)$ times plus construction, with ${}^{\leqslant n}L : {}^{\leqslant n}\mathbf{pStk} \to {}^{\leqslant n}\mathbf{Stk} \to {}^{\leqslant n}\mathbf{pStk}$ preserving k-truncated objects.

1.1.5. Schemes and Artin stacks

Definition 1.1.5.1. We define an object $Z \in \mathbf{pStk}$ to be a scheme if the following condition holds:
(1) Z satisfies Zariski descent.
(2) The diagonal $\Delta : Z \to Z \times Z$ is affine schematic, and for every $T \in (\mathbf{Aff})_{/Z\times Z}$, we have the induced map $(T \times_{Z\times Z} Z)_{\mathrm{cl}} \to T_{\mathrm{cl}}$ a closed immersion.
(3) There is a family of affine schemes $f_i : S_i \to Z$ such that each f_i is an open embedding, and the family is jointly surjective, i.e., $(f_i)_{\mathrm{cl}} : (S_i)_{\mathrm{cl}} \to Z_{\mathrm{cl}}$ is jointly surjective.

Definition 1.1.5.2. We then inductively define the notion of k-Artin stacks.
(1) We define an 0-Artin stack to be

Definition 1.1.5.3. We say an Artin stack is quasi-compact if it admit a smooth affine atlas.

We say a morphism of prestacks $\mathcal{X} \to \mathcal{Y}$ is quasi-compact if its pullback along an affine scheme is a quasi-compact Artin stack.

Definition 1.1.5.4. We let ${}^{\leqslant n}\mathbf{Sch}_{\mathrm{lft}} := \mathbf{Sch} \cap {}^{\leqslant n}\mathbf{Stk}_{\mathrm{lft}}$ and $\mathbf{Sch}_{\mathrm{laft}} := \mathbf{Sch} \cap \mathbf{Stk}_{\mathrm{laft}}$, and define ${}^{\leqslant n}\mathbf{Sch}_{\mathrm{ft}}$ and $\mathbf{Sch}_{\mathrm{aft}}$ to be the intersection of these categories with the quasi-compact schemes (Definition 1.1.5.3).

1.2. Quasi-coherent sheaves

1.2.1. Basic properties

Definition 1.2.1.1. We define the category of quasi-coherent sheaves over affine schemes to be a functor

$$\mathrm{QCoh}_{\mathbf{Aff}} : \mathbf{Aff} = \mathbf{CAlg}(\mathbf{Vect}^{\leqslant 0})^{\mathrm{op}} \to \mathbf{DGCat}$$

defined to be the composition

$$\mathbf{CAlg}(\mathbf{Vect}^{\leqslant 0})^{\mathrm{op}} \to \mathbf{CAlg}(\mathbf{Vect})^{\mathrm{op}} \xrightarrow{\mathbf{Mod}} \mathbf{DGCat}$$

where the second functor sends a ring A to the category $\mathbf{Mod}_A$ of its $\mathbb{E}_\infty$-modules, and a map of rings $A \to B$ the restriction functor $\mathbf{Mod}_B \to \mathbf{Mod}_A$. The functor is obviously continuous.

Equivalently, we interpret the above functor as $S \mapsto \mathrm{QCoh}(S)$ for $S \in \mathbf{Aff}$, and the transition functors are given by $f : S \to S'$ inducing $f_* : \mathrm{QCoh}(S) \to \mathrm{QCoh}(S')$. For this reason, we sometimes write the above functor as QCoh_* to indicate that we are using $(-)_*$ for the transitions.

remark 1.2.1.1. In [GR], the authors used the category of $\mathbb{E}_1$-left modules instead. We can see that this is equivalent to the category of $\mathbb{E}_\infty$-modules via the theorem of Lurie ([Lur17], Corollary 4.5.1.5). We remark here that although the category of $\mathbb{E}_1$-left modules admit no tensor structure, the viewpoint of $\mathbb{E}_1$-left modules is also useful, for example, when considering Koszul duality of polynomial algebras.

Definition 1.2.1.2. By passing to left adjoints, we obtain a functor $\mathrm{QCoh}^*_{\mathbf{Aff}} : \mathbf{Aff}^{\mathrm{op}} \to \mathbf{DGCat}$, to be thought of as assigning a morphism $f : S \to S'$ to $f^* : \mathrm{QCoh}(S') \to \mathrm{QCoh}(S)$.

Definition 1.2.1.3. We define $\mathrm{QCoh}^*_{\mathbf{pStk}} : \mathbf{pStk}^{\mathrm{op}} \to \mathbf{DGCat}$ to be the right Kan extension of the functor $\mathrm{QCoh}^*_{\mathbf{Aff}} : \mathbf{Aff}^{\mathrm{op}} \to \mathbf{DGCat}$ along the Yoneda embedding $\mathbf{Aff}^{\mathrm{op}} \to \mathbf{pStk}^{\mathrm{op}}$.

Unwinding the definition of the limit, for any prestack $\mathcal{X}$, we can think an object of $\mathrm{QCoh}(\mathcal{X})$ as assinging each $S \to \mathcal{X}$ an element $\mathcal{F}_S \in \mathrm{QCoh}(S)$ for S affine, and such that $\mathcal{F}_S$ are compatible with pulling back along the transition maps.

Since the limit is taken in $\mathbf{DGCat}$, for $f : \mathcal{X} \to \mathcal{Y}$, there is a functor $f_* : \mathrm{QCoh}(\mathcal{X}) \to \mathrm{QCoh}(\mathcal{Y})$ right adjoint to $f^* : \mathrm{QCoh}(\mathcal{Y}) \to \mathrm{QCoh}(\mathcal{X})$.

Proposition 1.2.1.4. For $\mathcal{X} \in {}^{\leqslant n}\mathbf{pStk}$, we have

$$\mathrm{QCoh}(\mathcal{X}) = \left(\mathrm{RKE}_{{}^{\leqslant n}\mathbf{Aff}^{\mathrm{op}} \hookrightarrow {}^{\leqslant n}\mathbf{pStk}^{\mathrm{op}}} \mathrm{QCoh}\right)(\mathcal{X})$$

Proof: By the cofinality obvious. □

Proposition 1.2.1.5. For $\mathcal{X} \in {}^{\leqslant n}\mathbf{pStk}_{\mathrm{lft}}$, we have

$$\mathrm{QCoh}(\mathcal{X}) = \left(\mathrm{RKE}_{{}^{\leqslant n}\mathbf{Aff}_{\mathrm{ft}}^{\mathrm{op}} \hookrightarrow {}^{\leqslant n}\mathbf{pStk}_{\mathrm{ft}}^{\mathrm{op}}} \mathrm{QCoh}\right)(\mathcal{X})$$

Proof: Again by the cofinality obvious. □

Construction 1.2.1.6. However, the functor $\mathrm{QCoh}^* : \mathbf{pStk}^{\mathrm{op}} \to \mathbf{DGCat}$ is not convergent. Take $S = \operatorname{Spec} k[\eta]$ with η in cohomological degree -2. Then, we have $k[\eta]$ the free $\mathbb{E}_1$-algebra under 1 generator in degree -2. Thus, an $k[\eta]$-module is simply a k-module equipped with a degree -2 endomorphism. However, taking the limit gives a k-module equipped with a *nilpotent* degree -2 endomorphism.

Proposition 1.2.1.7. The functor $\mathrm{QCoh}^*_{\mathbf{Aff}} : \mathbf{Aff}^{\mathrm{op}} \to \mathbf{DGCat}$ satisfies flat descent. Hence, it satisfies ppf, étale, and Zariski descent.

Proof: Apply the standard faithfully flat descent and the technique of the proof in Lemma 1.1.4.11. □

We now construct a t-structure on $\mathrm{QCoh}(\mathcal{X})$.

Construction 1.2.1.8. Recall we have the theorem of Lurie ([Lur17], Proposition 1.2.1.16) that a t-structure on a stable ∞-category $\mathcal{C}$ is equivalent to the data of a localization functor $\tau^{\leqslant 0} : \mathcal{C} \to \mathcal{C}$ subject to certain conditions.

We first recall that for $S \in \mathbf{Aff}$, $\mathrm{QCoh}(S)$ inherits a natural t-structure inherited from the forgetful functor $\mathrm{QCoh}(S) \to \mathbf{Vect}_k$. Notice that the pullback functors over affine schemes are right t-exact, for the rings are all connective. Then, by the description of limit categories by Cartesian sections, we see that colimits in such categories are computed pointwise. In particular, since $\mathrm{QCoh}(S)^{\leqslant 0}$ is closed under colimits, the subcategory

$$\mathrm{QCoh}(\mathcal{X})^{\leqslant 0} \subset \mathrm{QCoh}(\mathcal{X})$$

spanned by object with $(\mathcal{F}_S)$ satisfying $\mathcal{F}_S \in \mathrm{QCoh}(S)^{\leqslant 0}$ for any S is closed under colimits. Thus, we have its right adjoint $\tau^{\leqslant 0} : \mathrm{QCoh}(\mathcal{X}) \to \mathrm{QCoh}(\mathcal{X})^{\leqslant 0}$. Then by verifying the Ext condition of the localization functor $\tau^{\geqslant 1} = \mathrm{cofib}(\tau^{\leqslant 0} \to \mathrm{id})$, we obtain a t-structure on $\mathrm{QCoh}(\mathcal{X})$.

In particular, by our construction we have f^* is right t-exact and f_* is left t-exact.

Corollary 1.2.1.9. $\mathrm{QCoh}^{*,\leqslant 0}$ satisfies flat descent.

Proof: By construction. □

Lemma 1.2.1.10. $\mathrm{QCoh}^{*,\geqslant 0} : \left(\mathbf{pStk}^{k\text{-Artin}}\right)^{\mathrm{op}} \to \mathbf{DGCat}$ satisfies flat descent.

Lemma 1.2.1.11. Let $f : X \to Y$ be a map of schemes almost of finite type. Then, we have the pushforward $f_* : \mathrm{QCoh}(X) \to \mathrm{QCoh}(Y)$ of finite tor-amplitude.

Proof: We have already that f_* is left t-exact. Now, for $\mathcal{F} \in \mathrm{QCoh}(X)^{\leqslant 0}$, we prove $f_*(\mathcal{F}) \in \mathrm{QCoh}(Y)^{\leqslant N}$ for some N. Firstly, we take $U \subset Y$ an affine open, and take $V = U \times_Y X$. Then, we have $\Gamma(U, f_*(\mathcal{F})) = \Gamma(V, \mathcal{F})$. We apply the cohomological devissage to compute this. Namely, we consider the spectral sequence associated to the Postnikov filtration of $\mathcal{F}$. The graded pieces are given by $\Gamma(V, \mathcal{H}^q(\mathcal{F})) = \Gamma(V_{\mathrm{cl}}, \mathcal{H}^q(\mathcal{F}))$, so we obtain a spectral sequence

$$E_2^{p,q} = H^p(V_{\mathrm{cl}}, \mathcal{H}^q(\mathcal{F})) \Longrightarrow H^{p+q}(\Gamma(V, \mathcal{F}))$$

Now, since $\mathcal{F} \in \mathrm{QCoh}(X)^{\leqslant 0}$, $\mathcal{H}^q(\mathcal{F}) = 0$ for $q > 0$, and since X is almost of finite type, V_{cl} is a finite dimensional Noetherian topological space, say of dimension d. Hence, $H^p(V_{\mathrm{cl}}, \mathcal{G}) = 0$ for all $p > d$. Hence, taking $N = d$, we have $H^n(\Gamma(V, F)) = 0$ for all $n > N$, hence $f_*(\mathcal{F}) \in \mathrm{QCoh}(Y)^{\leqslant d}$, where $d = \dim Y_{\mathrm{cl}}$. □

We have constructed the pushforward functor as above. In partiuclar, we notice that it is discontinuous, as it arise from a general right adjoint of a functor in $\mathbf{Pr}^L_{\mathrm{St}}$. However, there are some nice properties which we can deduce from and used to deduce the continuity of the pushforward functor. Namely, we have the following.

Proposition 1.2.1.12. Let $f : \mathcal{X}_1 \to \mathcal{X}_2$ be a morphism of prestacks, and consider the pullback diagram

$$\begin{array}{ccc} \mathcal{Y}_1 & \xrightarrow{g'} & \mathcal{X}_1 \\ {\scriptstyle f'}\downarrow & \ulcorner & \downarrow{\scriptstyle f} \\ \mathcal{Y}_2 & \xrightarrow[g]{} & \mathcal{X}_2 \end{array}$$

Then, if f is schematic and quasi-compact (recall the definitions in Definition 1.1.4.3 and Definition 1.1.5.3), we have the following property.
(1) $f_* : \mathrm{QCoh}(\mathcal{X}_1) \to \mathrm{QCoh}(\mathcal{X}_2)$ is continuous.
(2) The base change morphism $g^* \circ f_* \to f'_* \circ g'^*$ is an isomorphism.

Proof:

□

1.2.2. The symmetric monoidal structure of QCoh

We aim to give a right lax symmetric monoidal structure on QCoh^* in this subsection. Before this, we first prove a baby version of the theorem.

Proposition 1.2.2.1. Let $\mathcal{C}_1, \mathcal{C}_2 \in \mathbf{Pr}^L_{\mathrm{St}}$, and let T_1, T_2 be monads over $\mathcal{C}_1, \mathcal{C}_2$ respectively. Then, we have the functor $\mathbf{Alg}_{T_1}(\mathcal{C}_1) \otimes \mathbf{Alg}_{T_2}(\mathcal{C}_2) \to \mathbf{Alg}_{T_1 \otimes T_2}(\mathcal{C}_1 \otimes \mathcal{C}_2)$ an equivalence.

Proof: Notice that the forgetful functors U_{T_1} and U_{T_2} preserves geometric realizations, thus so does $U_{T_1} \otimes U_{T_2}$. By the Barr-Beck-Lurie theorem, to show

$$F_{T_1} \otimes F_{T_2} : \mathcal{C}_1 \otimes \mathcal{C}_2 \to \mathbf{Alg}_{T_1}(\mathcal{C}_1) \otimes \mathbf{Alg}_{T_2}(\mathcal{C}_2)$$

is a monadic left adjoint, it suffices to show $U_{T_1} \otimes U_{T_2}$ is conservative. This is because $\mathcal{C}_1 \otimes \mathcal{C}_2$ is generated by the image of $\mathcal{C}_1 \times \mathcal{C}_2$, by taking a presentation of $\mathcal{C}_i$ by sites.

Then, the monads induced by $F_{T_1} \otimes F_{T_2}$ and $F_{T_1 \otimes T_2}$ clearly coincide, thus identifying the corresponding algebra categories. □

Corollary 1.2.2.2. The functor $\mathrm{QCoh}^*_{\mathbf{Aff}} : \mathbf{Aff}^{\mathrm{op}} \to \mathbf{DGCat}$ admit a canonical symmetric monoidal structure, where $\mathbf{Aff}^{\mathrm{op}} \simeq \mathbf{CAlg}(\mathbf{Vect}^{\leqslant 0})$ is equipped with the coCartesian monoidal structure.

Construction 1.2.2.3. We now construct a right lax symmetric monoidal structure on the functor $\mathrm{QCoh}^* : \mathbf{pStk}^{\mathrm{op}} \to \mathbf{DGCat}$ using Corollary 1.2.2.2. Consider the presentation

$$\mathrm{QCoh}(\mathcal{X}) \otimes \mathrm{QCoh}(\mathcal{Y}) = \left(\lim_{S \to \mathcal{X}} \mathrm{QCoh}(S)\right) \otimes \left(\lim_{T \to \mathcal{Y}} \mathrm{QCoh}(T)\right)$$

and

$$\mathrm{QCoh}(\mathcal{X}\times\mathcal{Y}) = \lim_{S\to\mathcal{X}\times\mathcal{Y}} \mathrm{QCoh}(S)$$

However, the functor $\mathbf{Aff}^{\mathrm{op}}_{/\mathcal{X}} \times \mathbf{Aff}^{\mathrm{op}}_{/\mathcal{Y}} \to \mathbf{Aff}^{\mathrm{op}}_{/\mathcal{X}\times\mathcal{Y}}$ is cofinal, so

$$\mathrm{QCoh}(\mathcal{X}\times\mathcal{Y}) = \lim_{S\times T\to\mathcal{X}\times\mathcal{Y}} \mathrm{QCoh}(S)\otimes\mathrm{QCoh}(T)$$

and the desired $\mathrm{QCoh}(\mathcal{X})\otimes\mathrm{QCoh}(\mathcal{Y}) \to \mathrm{QCoh}(\mathcal{X}\times\mathcal{Y})$ is then given by exchanging limits with the tensor product.

We denote the functor $\mathrm{QCoh}(\mathcal{X})\otimes\mathrm{QCoh}(\mathcal{Y}) \to \mathrm{QCoh}(\mathcal{X}\times\mathcal{Y})$ by the exterior tensor, whose precomposition with $\mathrm{QCoh}(\mathcal{X})\times\mathrm{QCoh}(\mathcal{Y}) \to \mathrm{QCoh}(\mathcal{X})\otimes\mathrm{QCoh}(\mathcal{Y})$ denoted by $\mathcal{F}_1\otimes\mathcal{F}_2 \mapsto \mathcal{F}_1\boxtimes\mathcal{F}_2$.

We also call the exterior tensor map the categorical Künneth formula map, and say the categorical Künneth formula holds if the above map is an equivalence of categories.

Corollary 1.2.2.4. Since taking tensor products with dualizable categories commutes with taking limits, the categorical Künneth formula holds for the stacks where one has QCoh dualizable.

Construction 1.2.2.5. We now apply Construction 1.2.2.3 to give a symmetric monoidal structure on the category $\mathrm{QCoh}(\mathcal{X})$ for any prestack $\mathcal{X}$. We define

$$\mathcal{F}_1\otimes\mathcal{F}_2 = \Delta^*_{\mathcal{X}}(\mathcal{F}_1\boxtimes\mathcal{F}_2)$$

Thus, the pullback functor is naturally symmetric monoidal. The pushforward functor, as its right adjoint, is naturally right lax symmetric monoidal. This in particular yields a projection formula map

$$\mathcal{F}\otimes f_*(\mathcal{G}) \to f_*(f^*(\mathcal{F})\otimes\mathcal{G})$$

for any $f:\mathcal{X}\to\mathcal{Y}$ and any $\mathcal{F}\in\mathrm{QCoh}(\mathcal{Y})$ and $\mathcal{G}\in\mathrm{QCoh}(\mathcal{X})$. We say the projection formula holds if the above map is an isomorphism.

Proposition 1.2.2.6. The projection formula for QCoh holds for affine schematic morphisms $f:\mathcal{X}\to\mathcal{Y}$.

Proof:

□

todo 1.2.2.1. Finish the proofs of the properties of affine schematic morphisms.

We then consider a stronger version of the categorical Künneth formula for QCoh. Namely, we consider when the following map is an equivalence.

$$\mathrm{QCoh}(\mathcal{X})\otimes_{\mathrm{QCoh}(\mathcal{Z})}\mathrm{QCoh}(\mathcal{Y}) \to \mathrm{QCoh}(\mathcal{X}\times_{\mathcal{Z}}\mathcal{Y})$$

It turns out that this is related to the rigidity of QCoh.

todo 1.2.2.2. Finish this part.

1.3. Ind-coherent sheaves

We then move on to the topic of ind-coherent sheaves. For a brief summary, recall that we have that for smooth classical schemes, the coherent sheaves can be identified with the perfect sheaves (Lemma 1.3.1.4). However, when S is singular, the two categories differ, however slightly. We have that they always agree on the eventually coconnective parts (todo 2.1). This suggests that coherent sheaves can detect the singularities of the scheme. Then, the category of coherent sheaves is not presentable, since it does not contain infinite direct sums. The way to resolve this is to consider the ind-coherent sheaves, which is a certain ind-completion of coherent sheaves. Going back to the topic of singularities, we can define the singular support of ind-coherent sheaves, which contains the data of where this ind-coherent sheaf is non-perfect. In particular, the quasi-coherent sheaves can be identified with the ind-coherent sheaves with zero singular support.

A second reason for considering ind-coherent sheaves is that the functoriality for ind-coherent sheaves are better than the ones for quasi-coherent sheaves. We will see this in todo 2.1.

1.3.1. Definitions

Definition 1.3.1.1. For $X \in \mathbf{Sch}_{\mathrm{aft}}$, we define $\mathrm{Coh}(X)$ to be the category of bounded complexes with cohomologies lying in $\mathrm{Coh}(X_{\mathrm{cl}})$. We define $\mathrm{IndCoh}(X)$ to be the ind-completion $\mathrm{Ind}(\mathrm{Coh}(X))$.

On the level of affine schemes, for $A \in \mathbf{CAlg}_{\mathrm{aft}}^{\leqslant 0}$ we have $\mathrm{Coh}(\operatorname{Spec} A)$ the subcategory of $\mathbf{Mod}_A$ given by objects M with bounded cohomologies and such that $H^i(M)$ is finitely generated (equivalent to finitely presented) as an $H^0(A)$-module.

By definition, there is a comparison map $\Psi_X : \mathrm{IndCoh}(X) \to \mathrm{QCoh}(X)$ given by ind-extending the inclusion $\mathrm{Coh}(X) \hookrightarrow \mathrm{QCoh}(X)$.

Definition 1.3.1.2. Let $A \in \mathbf{CAlg}_{\mathrm{aft}}^{\leqslant 0}$. We say $M \in \mathbf{Mod}_A$ is almost perfect if $M \in \mathbf{Mod}_A^{\leqslant k}$ for some k, and $\tau^{\geqslant n}(M)$ is compact in $\mathbf{Mod}_A^{[n,k]}$ for all n.

Theorem 1.3.1.3 ([Lur17], Proposition 7.2.4.17). Let $A \in \mathbf{CAlg}_{\mathrm{aft}}^{\leqslant 0}$ and $M \in \mathbf{Mod}_A$. Then, M is almost perfect iff the following conditions holds:
(1) $M \in \mathbf{Mod}_A^-$. That is, $M \in \mathbf{Mod}_A^{\leqslant k}$ for some k.
(2) For every n we have $H^n(M)$ finitely generated as an $H^0(A)$-module.

Thus, a coherent module is by definition an almost perfect module which is also eventually coconnective. We will use this later on.

Lemma 1.3.1.4. When X is a smooth classical scheme finite type over a field, Ψ_X is an equivalence of categories.

Proof: It suffices to prove that $\mathrm{Coh}(X) = \mathrm{Perf}(X)$. Notice that since X is Noetherian, the property of coherence and perfectness is local. Hence, we may assume that $X = \operatorname{Spec} A$ where A is a Noetherian regular ring of finite dimension. Hence, A is of finite global dimension. Thus every

finitely generated (i.e. coherent) A-module admit a finite length resolution. Thus we have obtained the result. □

Definition 1.3.1.5. We say X is eventually coconnective if $\mathcal{O}_X \in \mathrm{Coh}(X)$.

Lemma 1.3.1.6. For eventually coconnective X, the functor Ψ_X admits a fully faithful left adjoint, which we will denote by Ξ_X.

Proof: If $\mathcal{O}_X \in \mathrm{Coh}(X)$, then we have a fully faithful functor $\mathrm{Perf}(X) \hookrightarrow \mathrm{Coh}(X)$. Then, for such schemes, we have $\mathrm{QCoh}(X) = \mathrm{Ind}(\mathrm{Perf}(X))$, so Ξ_X is obtained by ind-extending the inclusion. Hence, $\Psi_X \circ \Xi_X$ is the ind-extension of the inclusion $\mathrm{Perf}(X) \hookrightarrow \mathrm{QCoh}(X)$, which is clearly the identity. The left adjoint property is obvious. □

We now construct a t-structure on the category $\mathrm{IndCoh}(X)$.

Lemma 1.3.1.7. Let $\mathcal{C}_0 \in \mathbf{DGCat}^{\text{non-cocmpl}}$ be a non-cocomplete DG category, endowed with a t-structure. Then, we have $\mathcal{C} = \mathrm{Ind}(\mathcal{C}_0) \in \mathbf{DGCat}$ admit a natural t-structure, with the following properties:
(1) The inclusion $\mathcal{C}_0 \hookrightarrow \mathcal{C}$ is t-exact.
(2) The subcategories $\mathcal{C}^{\leqslant 0}$ and $\mathcal{C}^{\geqslant 0}$ are compactly generated under filtered colimits by $\mathcal{C}_0^{\leqslant 0}$ and $\mathcal{C}_0^{\geqslant 0}$.
(3) Let $\mathcal{D}$ be another DG category with a t-structure, such that $\mathcal{D}^{\geqslant 0}$ is closed under filtered colimits. Then, a continuous functor $F : \mathcal{C} \to \mathcal{D}$ is t-exact iff $F|_{\mathcal{C}_0}$ is.

Proof: The existence of the t-structure is an application of [Lur17], Proposition 1.2.1.16. For (3), if $F|_{\mathcal{C}_0}$ is t-exact, we clearly have F preserves both connective and coconnective objects, since both are closed under taking filtered colimits. □

Via Lemma 1.3.1.7, we may define a t-structure on $\mathrm{IndCoh}(X)$.

Corollary 1.3.1.8. The t-structure on $\mathrm{IndCoh}(X)$ makes Ψ_X t-exact, and is compatible with filtered colimits.

Proposition 1.3.1.9. The functor $\Psi_X : \mathrm{IndCoh}\,(X)^{\geqslant 0} \to \mathrm{QCoh}(X)^{\geqslant 0}$ is an equivalence of categories. Hence, we have $\mathrm{IndCoh}(X)^+ = \mathrm{QCoh}(X)^+$.

Proof: It suffices to show that $\mathrm{QCoh}(X)^{\geqslant 0}$ is compactly generated by $\mathrm{Coh}(X)^{\geqslant 0}$ under filtered colimits. This is done by first passing to modules, and use that coherent sheaves are almost perfect. □

Lemma 1.3.1.10. We have $\mathrm{Coh}(X) = \mathrm{IndCoh}(X)^{\omega}$.

Proof: Firstly note that $\mathrm{IndCoh}(X)^\omega$ is the Karoubi completion of $\mathrm{Coh}(X)$. Then, for $\mathcal{F} \in \mathrm{IndCoh}(X)^\omega$, assume it is a direct summand (retract) of $\mathcal{F}' \in \mathrm{Coh}(X)$. Then we have $\Psi_X(\mathcal{F})$ is a direct summand of $\Psi_X(\mathcal{F}')$. But since $\mathrm{Coh}(X)$ is Karoubi complete in $\mathrm{QCoh}(X)$, we may assume $\Psi_X(\mathcal{F}) = \Psi_X(\mathcal{F}')$. Then by definition, $\mathcal{F}, \mathcal{F}' \in \mathrm{IndCoh}(X)^\omega \subset \mathrm{IndCoh}(X)^+ \simeq \mathrm{QCoh}(X)^+$, we have $\mathcal{F} = \mathcal{F}'$. □

Construction 1.3.1.11. We now construct an action of $\mathrm{QCoh}(X)$ on $\mathrm{IndCoh}(X)$, such that $\Psi_X : \mathrm{IndCoh}(X) \to \mathrm{QCoh}(X)$ is a map of $\mathrm{QCoh}(X)$-modules. For the action, notice that $\mathrm{Perf}(X) \subset \mathrm{Coh}(X)$ gives an action, whose ind-extension is the desired action. Moreover, we claim that this action is unique because the action of $\mathrm{Perf}(X)$ on $\mathrm{IndCoh}(X)$ must preserves compact objects, which is by dualizability of $\mathrm{Perf}(X)$.

Lemma 1.3.1.12. Suppose $X \in \mathbf{Sch}_{\mathrm{aft}}$ and $\mathcal{E} \in \mathrm{QCoh}(X)^b$, with $\mathcal{E} \otimes - : \mathrm{QCoh}(X) \to \mathrm{QCoh}(X)$ left t-exact up to finite shift. Then, the functor

$$\mathcal{E} \otimes - : \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(X)$$

is also left t-exact up to finite shift.

We temporarily postpone the proof (here) of this lemma to the following section, after we have developed certain functorialities of IndCoh.

1.3.2. The usual pushforward and pullback functors

Unlike the QCoh, we need to construct the pullback and pushforward functors by ourselves, and prove that they satisfy a good functoriality.

Construction 1.3.2.1. Let $f : X \to Y$ be a morphism in $\mathbf{Sch}_{\mathrm{aft}}$. We construct a functor

$$f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(Y) \in \mathbf{DGCat}$$

making the following diagram commute:

$$\begin{array}{ccc} \mathrm{IndCoh}(X) & \xrightarrow{\Psi_X} & \mathrm{QCoh}(X) \\ \downarrow{\scriptstyle f_*^{\mathrm{IndCoh}}} & & \downarrow{\scriptstyle f_*} \\ \mathrm{IndCoh}(Y) & \xrightarrow[\Psi_Y]{} & \mathrm{QCoh}(Y) \end{array}$$

Moreover, we can show that the constructed functor f_*^{IndCoh} is left t-exact.

By definition, functors in **DGCat** must be continuous, so f_*^{IndCoh} is the ind-extension of the restriction to $\mathrm{Coh}(X)$. Then, the above commutative diagram is nothing but

$$\Psi_Y \circ f_*^{\mathrm{IndCoh}}|_{\mathrm{Coh}(X)} = f_*|_{\mathrm{Coh}(X)}$$

Since $f_* : \mathrm{QCoh}(X) \to \mathrm{QCoh}(Y)$ is left t-exact and $\mathrm{Coh}(X) \subset \mathrm{QCoh}(X)^+$, we have the image of the functor $\Psi_Y \circ f_*^{\mathrm{IndCoh}}$ contained in $\mathrm{QCoh}(Y)^+$. Hence, we may define $f_*^{\mathrm{IndCoh}}|_{\mathrm{Coh}(X)}$ by

$$f_*^{\mathrm{IndCoh}}|_{\mathrm{Coh}(X)} = \Psi_Y|^{-1}_{\mathrm{IndCoh}(Y)^+} \circ f_*|_{\mathrm{Coh}(X)}$$

and finally define f_*^{IndCoh} by ind-extending the above functor.

Moreover, we have $f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(Y)$ is a functor of $\mathrm{QCoh}(Y)$-modules.

Lemma 1.3.2.2. Let $f : X \to Y$ be a morphism in $\mathbf{Sch}_{\mathrm{aft}}$. Then, f_*^{IndCoh} is right t-exact up to a shift, hence is of finite tor-amplitude.

Proof: This follows from Lemma 1.2.1.11 □

Proposition 1.3.2.3. The above construction upgrades to a functor

$$\mathrm{IndCoh}_{\mathbf{Sch}_{\mathrm{aft}}} : \mathbf{Sch}_{\mathrm{aft}} \to \mathbf{DGCat}$$

equipped with a natural transformation

$$\Psi : \mathrm{IndCoh}_{\mathbf{Sch}_{\mathrm{aft}}} \to \mathrm{QCoh}_{\mathbf{Sch}_{\mathrm{aft}}}$$

such that at the level of objects and 1-morphisms we obtain the above construction.

The construction of the full functorial structure is not direct because we need to specify the higher structures. However, we do not need to determine them by force, but via a clever construction of an auxiliary ∞-category. We will use similar techniques to obtain further functorialities of IndCoh.

Definition 1.3.2.4. We define a 1-fully faithful functor of ∞-categories to be a functor $F : \mathcal{C} \to \mathcal{D}$ such that $\mathrm{Hom}_{\mathcal{C}}(c_1, c_2) \to \mathrm{Hom}_{\mathcal{D}}(F(c_1), F(c_2))$ is an inclusion of connected components of spaces. In this case, we say $\mathcal{C}$ is a 1-full subcategory of $\mathcal{D}$.

We first give a precise formulation of our current status of the functoriality to be constructed.

Definition 1.3.2.5. Let $\mathcal{I}$ and $\mathcal{C}$ be ∞-categories. We define an assignment F from $\mathcal{I}$ to $\mathcal{C}$ to be a map of sets $\pi_0(\mathcal{I}^{\simeq}) \to \pi_0(\mathcal{C}^{\simeq})$, denoted by $i \rightsquigarrow F(i)$. We say it can be extended to a functor if there is a functor $F' : \mathcal{I} \to \mathcal{C}$ such that $F'(i) = F(i)$ for all $i \in \pi_0(\mathcal{I}^{\simeq})$. We abuse notations and denote this functor F' also by F.

Lemma 1.3.2.6. Let $T : \mathcal{C} \to \mathcal{D}$ be a 1-fully faithful functor of ∞-categories. Let $\mathcal{I}$ be another ∞-category, and let $i \rightsquigarrow F'(i)$ be an assignment from $\mathcal{I}$ to $\mathcal{C}$ such that $i \rightsquigarrow T \circ F'(i)$ can be extended to a functor $F : \mathcal{I} \to \mathcal{D}$. Then, suppose that for every $\alpha \in \mathrm{Hom}_{\mathcal{I}}(i_1, i_2)$, $F(\alpha) \in \mathrm{Hom}_{\mathcal{D}}(F(i_1), F(i_2))$ lies in the image of T. Then, there exists a unique extension of F' to a functor.

Proof: This is obvious. □

Lemma 1.3.2.7. We keep the settings as in the previous lemma. Now, let F_1' and F_2' be two functors from $\mathcal{I}$ to $\mathcal{C}$. Let $i \rightsquigarrow \psi_i' \in \mathrm{Hom}_{\mathcal{C}}(F_1'(i), F_2'(i))$ be an assignment from $\mathcal{I}$ to $\mathrm{Fun}(\Delta^1, \mathcal{C})$. Assume $i \rightsquigarrow T(\psi_i') \in \mathrm{Hom}_{\mathcal{D}}(F_1(i), F_2(i))$ can be extended to a functor $\psi : \mathcal{I} \to \mathrm{Fun}(\Delta^1, \mathcal{D})$, then there exists a unique extension of ψ' to a functor $\psi' : \mathcal{I} \to \mathrm{Fun}(\Delta^1, \mathcal{C})$.

Proof: Also obvious. □

Construction 1.3.2.8. We define the category $\mathbf{DGCat}^+$ to be the 1-full subcategory of $\mathbf{DGCat}^{\text{non-cocmpl}}$, spanned by objects that are non-cocomplete DG categories $\mathcal{C}$ with the following properties:
(1) It is equipped with a t-structure, and $\mathcal{C} = \mathcal{C}^+$.
(2) $\mathcal{C}^{\geqslant 0}$ has all filtered colimits and the inclusion $\mathcal{C}^{\geqslant 0} \hookrightarrow \mathcal{C}$ commutes with filtered colimits.

We take the 1-morphisms to be spanned by the functors $F : \mathcal{C}_1 \to \mathcal{C}_2$ left t-exact up to a finite shift, and $F|_{\mathcal{C}_1^{\geqslant 0}}$ preserves filtered colimits.

We define the category $\mathbf{DGCat}^t$ to be the 1-full subcategory of $\mathbf{DGCat}$, spanned by the objects that are DG categories $\mathcal{C}$ with the following properties:
(1) $\mathcal{C}$ is equipped with a t-structure, and is compactly generated by $\mathcal{C}^+$.
(2) $\mathcal{C}^{\geqslant 0}$ admits all filtered colimits.

We take the 1-morphisms to be spanned by the continuous functors left t-exact up to a finite shift.

Moreover, we can identify $\mathbf{DGCat}^t$ as a 1-full subcategory of $\mathbf{DGCat}^+$ by $\mathcal{C} \mapsto \mathcal{C}^+$.

Proof: We now prove Proposition 1.3.2.3. We set $\mathcal{I} = \mathbf{Sch}_{\text{aft}}$, $\mathcal{C} = \mathbf{DGCat}^t$, $\mathcal{D} = \mathbf{DGCat}$, $F = \mathrm{QCoh}_{\mathbf{Sch}_{\text{aft}}} : \mathbf{Sch}_{\text{aft}} \to \mathbf{DGCat}$, and the assignment F' to be $X \rightsquigarrow \mathrm{QCoh}(X) \in \mathbf{DGCat}^t$. Now, apply Lemma 1.3.2.6, we obtain a genuine functor $\mathrm{QCoh}^t_{\mathbf{Sch}_{\text{aft}}} : \mathbf{Sch}_{\text{aft}} \to \mathbf{DGCat}^t$. Then, we may compose into $\mathrm{QCoh}^+_{\mathbf{Sch}_{\text{aft}}} : \mathbf{Sch}_{\text{aft}} \to \mathbf{DGCat}^+$. We now set $\mathcal{C} = \mathbf{DGCat}^t$, $\mathcal{D} = \mathbf{DGCat}^+$, $F = \mathrm{QCoh}^+_{\mathbf{Sch}_{\text{aft}}}$, and the assignment $F' : X \rightsquigarrow \mathrm{IndCoh}(X) \in \mathbf{DGCat}^t$. Thus, again by Lemma 1.3.2.6, we obtain an extension $\mathrm{IndCoh}^t_{\mathbf{Sch}_{\text{aft}}} : \mathbf{Sch}_{\text{aft}} \to \mathbf{DGCat}^t$. Post-composing with the 1-fully faithful inclusion $\mathbf{DGCat}^t \hookrightarrow \mathbf{DGCat}$, we obtain the desired lift as a functor $\mathrm{IndCoh}_{\mathbf{Sch}_{\text{aft}}} : \mathbf{Sch}_{\text{aft}} \to \mathbf{DGCat}$. The natural transformation Ψ then is obtained by applying Lemma 1.3.2.7 to the obvious assignment, and the naturality condition used is ensured by Construction 1.3.2.1. □

We are not yet ready to define the category of ind-coherent sheaves for general prestacks. We will first construct a functor of $*$-pullbacks of ind-coherent sheaves for certain nice morphisms.

Definition 1.3.2.9. Let $f : X \to Y \in \mathbf{Sch}_{\text{aft}}$. We say f is eventually coconnective if $f^* : \mathrm{QCoh}(Y) \to \mathrm{QCoh}(X)$ sends $\mathrm{Coh}(Y)$ to $\mathrm{Coh}(X)$. The definition will remain unchanged if we replace $\mathrm{Coh}(X)$ above by $\mathrm{QCoh}(Y)^+$.

Lemma 1.3.2.10. f is eventually coconnective iff f^* is of finite tor-amplitude (as the definition Definition 1.1.1.2, and note f^* is naturally right t-exact, so here we use only left t-exactness up to a shift).

Proof: Obviously we have finite tor-amplitude implies eventual coconnectivity. For the converse, since the problem is Zariski local, we may factor f as $X \hookrightarrow Y \times \mathbb{A}^n \to Y$ by almost finite type condition. Then f is of finite tor-amplitude iff $X \hookrightarrow Y \times \mathbb{A}^n$ is, so we may assume f is a closed embedding.

Then, to test that f^* is of finite tor-amplitude, we compute $f^*\big(\mathrm{Coh}(Y)^\heartsuit\big)$. But then, since $\mathrm{Coh}(Y)^\heartsuit = \mathrm{Coh}(Y_{\mathrm{cl}})^\heartsuit$, we may assume Y is itself classical. Then since f is eventually coconnective, so is X. Now, since $f : X \to Y$ is affine schematic, so the projection formula holds, for $\mathcal{F} \in \mathrm{QCoh}(Y)$, we have

$$f_* f^*(\mathcal{F}) = f_*(\mathcal{O}_X) \otimes \mathcal{F}$$

But then f is a closed immersion so affine, and hence f_* is t-exact and conservative. So to prove f^* is of finite tor-amplitude, it suffices to prove $f_*(\mathcal{O}_X)$ is of finite tor-amplitude.

Now, notice that f also satisfy base change, so for any geometric point $s \hookrightarrow Y$, we have $s^* f_*(\mathcal{O}_X) = R\Gamma\big(\mathcal{O}_{X_s}\big)$ is eventually coconnective. Then, we prove that this implies $f_*(\mathcal{O}_X)$ is perfect. Let $E = f_*(\mathcal{O}_X)$. Suppose $s^*(E) \in \mathbf{Mod}_K^{[-n,m]}$, where $K = \mathcal{O}_s(s)$. We prove that on a neighborhood of s, $E \otimes -$ has amplitude $[-n, m]$. Firstly by shifting, we may assume $m = 0$. Then, we prove by induction.

The base case is $n = -1$, which says $s^* E = 0$. By derived Nakayama's lemma, we have the restriction of E to $\operatorname{Spec} \mathcal{O}_{Y,s}$ is 0. Hence, there is a neighborhood of s such that $E = 0$.

Then, we assume the induction hypothesis. We choose a finite rank free sheaf $P \to E$ inducing an isomorphism $H^0(s^* P) \xrightarrow{\simeq} H^0(s^* E)$. We let $F = \mathrm{fib}(P \to E)$. Hence, by the long exact sequence, we have $s^* F \in \mathbf{Mod}_K^{[-n+1,0]}$. Hence, F is of tor-amplitude $[-n+1, 0]$ on some neighborhood of s. Then on the same neighborhood, E is of tor-amplitude $[-n, 0]$. Finally, since Y is quasi-compact, we may find a uniform bound on the tor-amplitude of E. □

remark 1.3.2.1. Recall by [Sta26] 0658, being perfect is equivalent to being finite tor-amplitude and pseudo-coherent. So in fact, the sheaves appearing in the above proof are all perfect.

Definition 1.3.2.11. For $f : X \to Y$ eventually coconnective, we may ind-extend the functor $f^* : \mathrm{Coh}(Y) \to \mathrm{Coh}(X)$ and obtain $f^{\mathrm{IndCoh},*}$. In particular, it commutes with Ψ_X and f^* on coherent sheaves.

Lemma 1.3.2.12. $f^{\mathrm{IndCoh},*}$ is left adjoint to f_*^{IndCoh}.

Proof: Obviously we have for $\mathcal{F}_X \in \mathrm{Coh}(X)$ and $\mathcal{F}_Y \in \mathrm{Coh}(Y)$ an equivalence

$$\begin{aligned}&\mathrm{Hom}_{\mathrm{IndCoh}(X)}\big(f^{\mathrm{IndCoh},*}(\mathcal{F}_Y), \mathcal{F}_X\big) = \mathrm{Hom}_{\mathrm{QCoh}(X)}(f^*(\mathcal{F}_Y), \mathcal{F}_X)\\ &= \mathrm{Hom}_{\mathrm{QCoh}(Y)}(\mathcal{F}_Y, f_*(\mathcal{F}_{\mathcal{X}})) = \mathrm{Hom}_{\mathrm{IndCoh}(Y)}\big(\mathcal{F}_Y, f_*^{\mathrm{IndCoh}}(\mathcal{F}_X)\big)\end{aligned}$$

Then writing every object of IndCoh into filtered colimits of these compact generators, we are done. □

Corollary 1.3.2.13. $f^{\mathrm{IndCoh},*} : \mathrm{IndCoh}(Y) \to \mathrm{IndCoh}(X)$ is a map of $\mathrm{QCoh}(Y)$-modules.

Proof: Ind-extend the action of QCoh^{ω} on Coh. □

Proposition 1.3.2.14. $f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(Y)$ admit a left adjoint iff f is eventually coconnective.

Proof: Let $f^{\mathrm{IndCoh},L}$ denote the left adjoint. Since it preserves all colimits, it preserves compact objects. We let f^L denote the restriction $f^L : \mathrm{Coh}(X) \to \mathrm{Coh}(Y)$, and $f^{\mathrm{IndCoh},L}$ is its ind-extension. Hence, it suffices to prove that ??????? □

todo 1.3.2.1. Finish this proof

Corollary 1.3.2.15. By the same process as the proof of Proposition 1.3.2.3, we have the assignment $X \rightsquigarrow \mathrm{IndCoh}(X)$, $f \rightsquigarrow f^{\mathrm{IndCoh},*}$ extends to a functor

$$\mathrm{IndCoh}^*_{\mathbf{Sch}_{\mathrm{aft,ev\text{-}coconn}}} : \mathbf{Sch}^{\mathrm{op}}_{\mathrm{aft,ev\text{-}coconn}} \to \mathbf{DGCat}$$

1.3.3. Basic properties of the functors

We now establish some basic properties of the above constructed functors. We begin with the base change theorem.

Proposition 1.3.3.1. Consider the following Cartesian diagram in $\mathbf{Sch}_{\mathrm{aft}}$:

$$\begin{array}{ccc} X_1 & \xrightarrow{g_X} & X_2 \\ {\scriptstyle f_1}\downarrow & \ulcorner & \downarrow{\scriptstyle f_2} \\ Y_1 & \xrightarrow[g_Y]{} & Y_2 \end{array}$$

Assume f_2 is eventually coconnective. Then, the base change morphism

$$f_2^{\mathrm{IndCoh},*} \circ g_{Y,*}^{\mathrm{IndCoh}} \to g_{X,*}^{\mathrm{IndCoh}} \circ f_1^{\mathrm{IndCoh},*}$$

is an equivalence

Proof: It suffices to show that this is an equivalence for $\mathcal{F} \in \mathrm{Coh}(Y_1)$. In this case, both sides belong to $\mathrm{IndCoh}(X_2)^+$. Thus, it suffices to show that the above map is an equivalence after applying Ψ_{X_2}. Using compatibility of Ψ's with f_*^{IndCoh} and $f^{\mathrm{IndCoh},*}$, this now follows from Proposition 1.2.1.12. □

Corollary 1.3.3.2. The following version of projection formula holds. In this case, we do not need f even to be eventually coconnective.

$$\mathcal{E}_Y \otimes f_*^{\mathrm{IndCoh}} \mathcal{F}_X \xrightarrow{\simeq} f_*^{\mathrm{IndCoh}}(f^*(\mathcal{E}_Y) \otimes \mathcal{F}_X)$$

Proof: Follows from $f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(Y)$ is a morphism of $\mathrm{QCoh}(Y)$-modules, as in Construction 1.3.2.1. □

The projection formula for f_*^{IndCoh} and $f^{\mathrm{IndCoh},*}$ is much more difficult. For these, we need the Zariski descent for IndCoh.

Lemma 1.3.3.3. Suppose $j : X \hookrightarrow Y$ is an open embedding, then j_*^{IndCoh} is fully faithful.

Proof: We show the counit of the adjunction $j^{\mathrm{IndCoh},*} \circ j_*^{\mathrm{IndCoh}} \to \mathrm{id}$ is an equivalence. Note on both sides the functors are continuous, so it suffices to check that for $\mathcal{F} \in \mathrm{Coh}(X)$ we have $j^{*,\mathrm{IndCoh}} \circ j_*^{\mathrm{IndCoh}}(\mathcal{F}) \to \mathcal{F}$ is an equivalence. Since both sides are in IndCoh^+, applying Ψ_X to both sides it suffices to show $j^* \circ j_* \to \mathrm{id}$ is an equivalence, which is obvious. □

Corollary 1.3.3.4. By Lemma 1.3.1.7 and Lemma 1.3.3.3, we have $j^{\mathrm{IndCoh},*}$ t-exact.

Proposition 1.3.3.5. Let $f : U \to X$ be a Zariski cover. Then, the functor

$$\mathrm{IndCoh}(X) \to \mathrm{Tot}(\mathrm{IndCoh}(U^\bullet))$$

is an equivalence.

Proof: By compactness of X, we immediately reduces to the following case:

$$X = U_1 \cup U_2, U_{12} = U_1 \cap U_2$$

we must show

$$\mathrm{IndCoh}(X) \to \mathrm{IndCoh}(U_1) \times_{\mathrm{IndCoh}(U_{12})} \mathrm{IndCoh}(U_2)$$

is an equivalence.

We first construct the right adjoint of this functor. We send the data

$$\left(\mathcal{F}_i \in \mathrm{IndCoh}(U_i), \mathcal{F}_{12} \in \mathrm{IndCoh}(U_{12}), \mathcal{F}_1, j_{12,1}^{\mathrm{IndCoh},*}(\mathcal{F}_1) = \mathcal{F}_{12} = j_{12,2}^{\mathrm{IndCoh},*}(\mathcal{F}_2)\right)$$

to the obvious ind-coherent sheaf

$$\ker\left(j_{1,*}^{\mathrm{IndCoh}}(\mathcal{F}_1) \oplus j_{2,*}^{\mathrm{IndCoh}}(\mathcal{F}_2) \to j_{12,*}^{\mathrm{IndCoh}}(\mathcal{F}_{12})\right)$$

By the base change theorem Proposition 1.3.3.1 and Lemma 1.3.3.3, the composition

$$\mathrm{IndCoh}(U_1) \times_{\mathrm{IndCoh}(U_{12})} \mathrm{IndCoh}(U_2) \to \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(U_1) \times_{\mathrm{IndCoh}(U_{12})} \mathrm{IndCoh}(U_2)$$

is an equivalence. Then, to show that

$$\mathrm{IndCoh}(X) \to \mathrm{IndCoh}(U_1) \times_{\mathrm{IndCoh}(U_{12})} \mathrm{IndCoh}(U_2) \to \mathrm{IndCoh}(X)$$

is an equivalence, it suffices to show that for $\mathcal{F} \in \mathrm{IndCoh}(X)$, we have

$$\ker\left(j_{1,*}^{\mathrm{IndCoh}} \circ j_1^{\mathrm{IndCoh},*}(\mathcal{F}) \oplus j_{2,*}^{\mathrm{IndCoh}} \circ j_2^{\mathrm{IndCoh},*}(\mathcal{F}) \to j_{12,*}^{\mathrm{IndCoh}} \circ j_{12}^{\mathrm{IndCoh},*}(\mathcal{F})\right)$$

is an equivalence. Again, since all functors are continuous, we may assume $\mathcal{F} \in \mathrm{Coh}(X)$, and again both sides are in $\mathrm{IndCoh}\,(X)^+$, so we may apply Ψ_X to both sides. In this case, it reduces to the Zariski descent of quasi-coherent sheaves, which is by Proposition 1.2.1.7. □

Corollary 1.3.3.6. Let $f : U \to X$ be a Zariski cover. Then, $\mathcal{F} \in \mathrm{IndCoh}(X)$ is in $\mathrm{IndCoh}(X)^{\geqslant 0}$ or $\mathrm{IndCoh}(X)^{\leqslant 0}$ iff $f^{\mathrm{IndCoh},*}(\mathcal{F})$ does.

Proof: For only if, we have $j^{\mathrm{IndCoh},*}$ t-exact by Corollary 1.3.3.4. For the other direction, assume $f^{\mathrm{IndCoh},*}(\mathcal{F}) \in \mathrm{IndCoh}(U)^{\leqslant 0}$, apply Ψ_U, and we obtain the result from Proposition 1.2.1.7. Assume $f^{\mathrm{IndCoh},*}(\mathcal{F}) \in \mathrm{IndCoh}(U)^{\geqslant 0}$, the assertion follows from the construction of the inverse functor as in Proposition 1.3.3.5. □

Proof: We may now prove Lemma 1.3.1.12. By Lemma 1.3.1.7, it suffices to prove that $\mathcal{E} \otimes -$ sends $\mathrm{Coh}(X)^{\geqslant 0}$ to $\mathrm{IndCoh}(X)^{\geqslant -n}$. To show this, it suffices to prove $\mathrm{Coh}(X)^{\heartsuit}$ is mapped to $\mathrm{IndCoh}(X)^{\geqslant -n}$. We consider the closed embedding $i : X_{\mathrm{cl}} \hookrightarrow X$. Note $i_*^{\mathrm{IndCoh}} = i_* : \mathrm{Coh}(X_{\mathrm{cl}})^{\heartsuit} \to \mathrm{Coh}(X)^{\heartsuit}$ induces an equivalence of categories. By the projection formula above, we have

$$\mathcal{E} \otimes i_*^{\mathrm{IndCoh}}(\mathcal{F}') \simeq i_*^{\mathrm{IndCoh}}(i^*(\mathcal{E}) \otimes \mathcal{F}')$$

and notice that $i^*(\mathcal{E})$ is bounded of the same tor amplitude as $\mathcal{E}$, and i_*^{IndCoh} is t-exact, we may assume that X is classical. Moreover, by Corollary 1.3.3.6, we may assume that X is affine. In this case, the assumption that $\mathcal{E}$ is bounded of tor amplitude $[-n, 0]$ is equivalent that $\mathcal{E}$ has a flat resolution of length at most n. Again by cohomological devissage, it suffices to prove for $\mathcal{E}$ a flat module. In this case, $\mathcal{E} \otimes - : \mathrm{Coh}(X) \to \mathrm{Coh}(X)$ is obviously t-exact, and by Lemma 1.3.1.7 so is $\mathcal{E} \otimes - : \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(X)$. □

Corollary 1.3.3.7. The following version of projection formula holds.

$$f_*(\mathcal{E}_X) \otimes \mathcal{F}_Y \xrightarrow{\simeq} f_*^{\mathrm{IndCoh}}\left(\mathcal{E}_X \otimes f^{\mathrm{IndCoh},*}(\mathcal{F}_Y)\right)$$

Proof: It suffices to construct the natural transformation and prove that it is an equivalence on the level of functors $\mathrm{QCoh}(X)^{\omega} \otimes \mathrm{Coh}(Y) \to \mathrm{IndCoh}(Y)$, and both sides are ind-extended from here. Note that after post-composing Ψ_Y, this is just the ordinary projection formula for QCoh, which is an equivalence. Then, we have $f^{\mathrm{IndCoh},*}(\mathcal{F}_Y) = f^*(\mathcal{F}_Y) \in \mathrm{Coh}(X)$, so by Construction 1.3.1.11 we have $\mathcal{E}_X \otimes f^{\mathrm{IndCoh},*}(\mathcal{F}_Y) \in \mathrm{Coh}(X)$. Finally, by Construction 1.3.2.1 we have f_*^{IndCoh} preserves coherent sheaves, so we obtain

$$f_*^{\mathrm{IndCoh}}\left(\mathcal{E}_X \otimes f^{\mathrm{IndCoh},*}(\mathcal{F}_Y)\right) \in \mathrm{Coh}(Y) \subset \mathrm{IndCoh}(Y)^+$$

Hence, it suffices to prove

$$f_*(\mathcal{E}_X) \otimes \mathcal{F}_Y \in \mathrm{IndCoh}\,(Y)^+$$

where Ψ_Y is an equivalence.

By the analogous steps of the proof in Lemma 1.3.2.10, we have $f_*(\mathcal{E}_X)$ finite tor amplitude. By Lemma 1.3.1.12, the desired conclusion follows. □

Then, we investigate more properties of the action of QCoh on IndCoh. Namely, we aim to prove the following property.

Proposition 1.3.3.8. We have a fully faithful functor

$$\left(\mathrm{id}_{\mathrm{QCoh}(X)} \otimes f^{\mathrm{IndCoh},*}\right) : \mathrm{QCoh}(X) \otimes_{\mathrm{QCoh}(Y)} \mathrm{IndCoh}(Y) \to \mathrm{IndCoh}(X)$$

Proof: The left hand side is compactly generated by objects of the form $\mathcal{E}_X \otimes \mathcal{F}_Y$, where $\mathcal{E}_X \in \mathrm{QCoh}(X)^\omega$ and $\mathcal{F}_Y \in \mathrm{Coh}(Y)$. Moreover, above functor preserves compact objects. We thus suffices to show for compact $\mathcal{E}_X^1, \mathcal{E}_X^2, \mathcal{F}_Y^1, \mathcal{F}_Y^2$, the following equivalence

$$\mathrm{Hom}_{\mathrm{QCoh}(X)\otimes_{\mathrm{QCoh}(Y)}\mathrm{IndCoh}(Y)}(\mathcal{E}_X^1 \otimes \mathcal{F}_Y^1, \mathcal{E}_X^2 \otimes \mathcal{F}_Y^2) \xrightarrow{\simeq}$$
$$\mathrm{Hom}_{\mathrm{IndCoh}(X)}(\mathcal{E}_X^1 \otimes f^{\mathrm{IndCoh},*}(\mathcal{F}_Y^1), \mathcal{E}_X^2 \otimes f^{\mathrm{IndCoh},*}(\mathcal{F}_Y^2))$$

By dualizability of $\mathcal{E}_X^1$, by replacing $\mathcal{E}_X^2$ by tensoring it with $(\mathcal{E}_X^1)^\vee$, it suffices to show

$$\mathrm{Hom}_{\mathrm{QCoh}(X)\otimes_{\mathrm{QCoh}(Y)}\mathrm{IndCoh}(Y)}(\mathcal{F}_Y^1, \mathcal{E}_X \otimes \mathcal{F}_Y^2) \xrightarrow{\simeq} \mathrm{Hom}_{\mathrm{IndCoh}(X)}(f^{\mathrm{IndCoh},*}(\mathcal{F}_Y^1), \mathcal{E}_X \otimes f^{\mathrm{IndCoh},*}(\mathcal{F}_Y^2))$$

Since $f^{\mathrm{IndCoh},*}$ is left adjoint to f_*^{IndCoh}, it suffices to show that

$$\mathrm{Hom}_{\mathrm{QCoh}(X)\otimes_{\mathrm{QCoh}(Y)}\mathrm{IndCoh}(Y)}(\mathcal{F}_Y^1, \mathcal{E}_X \otimes \mathcal{F}_Y^2) \xrightarrow{\simeq} \mathrm{Hom}_{\mathrm{IndCoh}(Y)}(\mathcal{F}_Y^1, f_*^{\mathrm{IndCoh}}(\mathcal{E}_X \otimes f^{\mathrm{IndCoh},*}(\mathcal{F}_Y^2)))$$

By Corollary 1.3.3.7, we have

$$f_*^{\mathrm{IndCoh}}(\mathcal{E}_X \otimes f^{\mathrm{IndCoh},*}(\mathcal{F}_Y^2)) = f_*(\mathcal{E}_X) \otimes \mathcal{F}_Y^2$$

and the rest is by definition. □

Corollary 1.3.3.9. The map in Proposition 1.3.3.8 is an equivalence when f is an open immersion.

Proof: The essential image of the functor clearly generates the target category. □

1.3.4. The exceptional pullback functor

Definition 1.3.4.1. Let $f : X \to Y \in \mathbf{Sch}_{\mathrm{aft}}$ be a functor. We say f is proper (or closed embedding) if f_{cl} is so.

Lemma 1.3.4.2. If $f : X \to Y$ is proper, then the functor $f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(Y)$ preserves compact objects. That is, it sends $\mathrm{Coh}(X)$ to $\mathrm{Coh}(Y)$.

Proof: By Construction 1.3.2.1, it suffices to show that $f_* : \mathrm{QCoh}(X) \to \mathrm{QCoh}(Y)$ sends $\mathrm{Coh}(X)$ to $\mathrm{Coh}(Y)$. By definition, for f a closed embedding this obviously holds. Then, it suffices to prove

f_* maps $\mathrm{Coh}(X)^\heartsuit$ to $\mathrm{Coh}(Y)$. Then, noticing $\mathrm{Coh}(X)^\heartsuit = \mathrm{Coh}(X_{\mathrm{cl}})^\heartsuit$, we may assume X is classical. Then, the map $X \to Y$ factors through Y_{cl}, and since $Y_{\mathrm{cl}} \hookrightarrow Y$ is a closed embedding, we may assume that Y is also classical.

Thus, we are reduced to proving that for a proper map of classical schemes of finite type $f : X \to Y$ over a field (hence Noetherian), we have $R^i f_*(\mathcal{F})$ coherent for any coherent $\mathcal{F}$. This is [Sta26] 02O5.

We will recall and use [Sta26] 01YI. Let X be a Noetherian scheme and $\mathcal{P}$ a property for coherent sheaves over X. Assume that $\mathcal{P}$ has the two-out-of-three property for any exact sequence, and for every integral closed subscheme $Z \subset X$ with generic point ξ, there is a coherent sheaf $\mathcal{G} \in \mathrm{Coh}(X)$ satisfying the following properties:
(1) it is supported on Z,
(2) $\mathcal{G}_\xi$ is annihilated by $\mathfrak{m}_\xi$,
(3) $\dim_{\kappa(\xi)} \mathcal{G}_\xi = 1$
(4) $\mathcal{G}$ has property $\mathcal{P}$

then $\mathcal{P}$ holds for any coherent sheaf over X. The proof of this lemma is basically a version of Noetherian induction.

Now, we set $\mathcal{P}$ to be $R^i f_*(\mathcal{F})$ is coherent for all i. We verify the properties above. Firstly, $\mathcal{P}$ satisfy the two-out-of-three for exact sequences is obvious by the long exact sequence of cohomology. Then, take $Z \subset X$ an integral closed subscheme. We wish to find a coherent sheaf satisfying the above properties. We let $g : Z \to Y$ denote the restriction of f. We will actually construct a coherent sheaf $\mathcal{G}$ over Z such that $g_*(\mathcal{G})$ has no higher images, $\mathcal{G}_\xi$ is 1-dimensional, and $g_*(\mathcal{C})$ is coherent. Then, pushing $\mathcal{G}$ to X gives a coherent sheaf with the desired properties. Thus everything reduces to finding such a coherent sheaf $\mathcal{G}$.

We apply Chow's lemma to $Z \to Y$. Namely, there is a proper Y-scheme map $\pi : Z' \twoheadrightarrow Z$, with a dense open $U \subset Z$ such that $\pi^{-1}(U) \to U$ is an isomorphism, and $i : Z' \hookrightarrow \mathbb{P}^n_Y$. By properness of π, the diagonal map $i' : Z' \hookrightarrow \mathbb{P}^n_Y \times_Y Z = \mathbb{P}^n_Z$ is a closed immersion. We take a line bundle

$$\mathcal{L} = i^*\big(\mathcal{O}_{\mathbb{P}^n_Y}(1)\big) = i'^*\big(\mathcal{O}_{\mathbb{P}^n_Z}(1)\big)$$

which is ample for both $g' : Z' \to S$ and $\pi : Z' \to Z$. By the equivalent characterization of ampleness, there is some m such that the higher direct images of $\mathcal{L}^{\otimes m}$ with respect to g' and π vanishes. We then set $\mathcal{G} = \pi_*(\mathcal{L}^{\otimes n})$. Note $\mathcal{G}$ is invertible over U since π is an isomorphism over U, we have $\mathcal{G}_\xi$ dimension 1 over $\kappa(\xi)$. The higher direct images vanishes by a computation of the Leray-Serre spectral sequence

$$E_2^{p,q} = R^p g_* \circ R^q \pi_*(\mathcal{L}^{\otimes n}) \Longrightarrow R^{p+q} g'_*(\mathcal{L}^{\otimes n})$$

we have the left hand side is non-zero iff $q = 0$, and the right hand side is non-zero iff $p + q = 0$, so $R^p g_*(\mathcal{G}) = 0$ for $p > 0$. Then, since g' is projective, $g_*(\mathcal{G}) = g'_*(\mathcal{L}^{\otimes n})$ is clearly coherent by a computation of the cohomology of line bundles over projective spaces. □

Definition 1.3.4.3. The above shows that for a proper map $f : X \to Y$, we have $f_*^{\mathrm{IndCoh}} : \mathrm{IndCoh}(X) \to \mathrm{IndCoh}(Y)$ is continuous, and its right adjoint preserves filtered colimits, hence preserves all colimits. Thus, f_*^{IndCoh} admit a continuous right adjoint $f^! : \mathrm{IndCoh}(Y) \to \mathrm{IndCoh}(X)$. This is a functor of $\mathrm{QCoh}(Y)$-modules.

By the general formalism of passing to right adjoints, we can define a functor

$$\mathrm{IndCoh}^!_{(\mathbf{Sch}_{\mathrm{aft}})_{\mathrm{proper}}} : (\mathbf{Sch}_{\mathrm{aft}})^{\mathrm{op}}_{\mathrm{proper}} \to \mathbf{DGCat}$$

By Proposition 1.2.1.12, the functor f_* admit a discontinuous right adjoint $f^{!,\mathrm{QCoh}}$.

Lemma 1.3.4.4. The square

$$\begin{array}{ccc} \mathrm{IndCoh}(X)^+ & \xleftarrow{f^!} & \mathrm{IndCoh}(Y)^+ \\ \Psi_X \downarrow & & \downarrow \Psi_Y \\ \mathrm{QCoh}(X)^+ & \xleftarrow[f^{\mathrm{QCoh},!}]{} & \mathrm{QCoh}(Y)^+ \end{array}$$

commutes.

Proof: Since Ψ_X and Ψ_Y are equivalences, and the functors f_*^{IndCoh} and f_* are intertwined by Ψ.□

The analogous fact is **not** true if we remove the eventually coconnectivity condition.

Proposition 1.3.4.5. Let

$$\begin{array}{ccc} X_1 & \xrightarrow{g_X} & X_2 \\ f_1 \downarrow & \ulcorner & \downarrow f_2 \\ Y_1 & \xrightarrow{g_Y} & Y_2 \end{array}$$

be a Cartesian diagram where f_i are proper maps. Then, we have the base change isomorphism

$$(g_X)^{\mathrm{IndCoh}}_* \circ f_1^! \simeq f_2^! \circ (g_Y)^{\mathrm{IndCoh}}_*$$

Proof: Note all functors involved are continuous, so it suffices to verify the equivalence for the compact objects $\mathcal{F} \in \mathrm{Coh}(Y_1)$. Then, the functors have images in the following diagram

$$\begin{array}{ccc} \mathrm{IndCoh}\,(X_1)^+ & \xrightarrow{(g_X)^{\mathrm{IndCoh}}_*} & \mathrm{IndCoh}\,(X_2)^+ \\ f_1^! \uparrow & & \uparrow f_2^! \\ \mathrm{IndCoh}\,(Y_1)^+ & \xrightarrow{(g_Y)^{\mathrm{IndCoh}}_*} & \mathrm{IndCoh}\,(Y_2)^+ \end{array}$$

To prove the above diagram commute, we first apply the equivalence $\Psi : \mathrm{IndCoh}^+ \to \mathrm{QCoh}^+$. Then, the equivalence follows from the base change for QCoh and passing to right adjoints

$$\begin{array}{ccc} \mathrm{QCoh}(X_1)^+ & \xleftarrow{g_X^*} & \mathrm{QCoh}(X_2)^+ \\ {\scriptstyle (f_1)_*}\downarrow & & \downarrow{\scriptstyle (f_2)_*} \\ \mathrm{QCoh}(Y_1)^+ & \xleftarrow{g_Y^*} & \mathrm{QCoh}(Y_2)^+ \end{array} \quad \begin{array}{c} \\ (-)^R \\ \rightsquigarrow \\ \end{array} \quad \begin{array}{ccc} \mathrm{QCoh}(X_1)^+ & \xrightarrow{(g_X)_*} & \mathrm{QCoh}(X_2)^+ \\ \uparrow{\scriptstyle f_1^{\mathrm{QCoh},!}} & & \uparrow{\scriptstyle f_2^{\mathrm{QCoh},!}} \\ \mathrm{QCoh}(Y_1)^+ & \xrightarrow{(g_Y)_*} & \mathrm{QCoh}(Y_2)^+ \end{array}$$

□

Proposition 1.3.4.6. Let

$$\begin{array}{ccc} X_1 & \xrightarrow{g_X} & X_2 \\ {\scriptstyle f_1}\downarrow & \ulcorner & \downarrow{\scriptstyle f_2} \\ Y_1 & \xrightarrow{g_Y} & Y_2 \end{array}$$

be a Cartesian diagram where f_i are proper and g_X, g_Y are eventually coconnective. Then, by the $(f_*^{\mathrm{IndCoh}}, f^!)$-adjunction and the $(*,*)$-base change for IndCoh (Proposition 1.3.3.1), we have a map

$$g_X^{\mathrm{IndCoh},*} \circ f_2^! \simeq f_1^! \circ g_Y^{\mathrm{IndCoh},*}$$

The above map is an equivalence.

Proof: Firstly, since both functors commutes with filtered colimits, it suffices to prove the equivalence for $\mathcal{F} \in \mathrm{Coh}(Y_2)$. Note that $f_i^!$ are left t-exact up to a finite shift by passing to its left adjoint and apply Lemma 1.3.2.2. Thus, since $g_X^{\mathrm{IndCoh},*}$ is also left t-exact, it suffices to prove the diagram

$$\begin{array}{ccc} \mathrm{IndCoh}(X_1)^+ & \xleftarrow{g_X^{*,\mathrm{IndCoh}}} & \mathrm{IndCoh}(X_2)^+ \\ {\scriptstyle f_1^!}\uparrow & & \uparrow{\scriptstyle f_2^!} \\ \mathrm{IndCoh}(Y_1)^+ & \xleftarrow{g_Y^{*,\mathrm{IndCoh}}} & \mathrm{IndCoh}(Y_2)^+ \end{array}$$

commutes.

Notice that the equivalence is Zariski local on Y_2, so we may assume that g_X and g_Y are affine. Since for f affine we have f_* is monadic, in particular it is conservative. Hence, it suffices to prove that

$$(g_X)_*^{\mathrm{IndCoh}} \circ g_X^{\mathrm{IndCoh},*} \circ f_2^! \simeq (g_X)_*^{\mathrm{IndCoh}} \circ f_1^! \circ g_Y^{\mathrm{IndCoh},*}$$

By Corollary 1.3.3.7, we have $(g_X)_*^{\mathrm{IndCoh}} \circ g_X^{\mathrm{IndCoh},*} \simeq g_{X,*}\left(\mathcal{O}_{X_1}\right) \otimes -$. But by definition,

$$g_{X,*}\left(\mathcal{O}_{X_1}\right) = g_{X,*} \circ f_1^*\left(\mathcal{O}_{Y_1}\right) = f_2^* \circ g_{Y,*}\left(\mathcal{O}_{Y_1}\right)$$

by Proposition 1.2.1.12.

Hence, we have

$$(g_X)_*^{\mathrm{IndCoh}} \circ g_X^{\mathrm{IndCoh},*} \circ f_2^!(\mathcal{F}) = f_2^*\left(g_{Y,*}\left(\mathcal{O}_{Y_1}\right)\right) \otimes f_2^!(\mathcal{F}) = f_2^!\left(g_{Y,*}\left(\mathcal{O}_{Y_1}\right) \otimes \mathcal{F}\right)$$

by linearity of $f_2^!$ as QCoh-modules.

On the other hand, we have

$$(g_X)_*^{\mathrm{IndCoh}} \circ f_1^! \circ g_Y^{\mathrm{IndCoh},*}(\mathcal{F}) = f_2^! \circ (g_Y)_*^{\mathrm{IndCoh}} \circ g_Y^{\mathrm{IndCoh},*}(\mathcal{F}) = f_2^!\Big(g_{Y,*}\Big(\mathcal{O}_{Y_1}\Big) \otimes \mathcal{F}\Big)$$

and unwinding definitions we obtain that the comparison map between this two is given by the identity. □

Then, we temporarily set $\mathbf{CAlg} = \mathbf{CAlg}(\mathbf{Sp})$, and give some descent results.

Lemma 1.3.4.7.

1.4. Application: 1-Cartier duality

The contents in this section may be treated over the more general base **Sp**. That is, we no longer suppose the base ring we are considering over is any field of characteristic 0, but over the sphere spectrum $\mathbb{S}$. The readers may assume that all the statements needed still holds.

2. Todo chapter

todo 2.1.

3. Gestalten

3.1. Higher presentable categories

3.1.1. Presentability

We first recall the following definitions of presentable categories. As usual, we fix an uncountable regular cardinal κ.

Definition 3.1.1.1. Let $\mathcal{C}$ be an $(\infty,1)$-category and $c \in \mathcal{C}$ an object. We say a diagram in $\mathcal{C}$ is κ-filtered if any subdiagram of size $< \kappa$ can be extended to a right cone.

We say c is κ-compact if $\mathrm{Hom}(c,-)$ commutes with κ-filtered colimits. We denote the κ-compact objects by $\mathcal{C}^\kappa$.

We say $\mathcal{C}$ is κ-presentable if there is a subset $S \subset \mathcal{C}$ of κ-compact objects such that $\mathcal{C}$ is generated by these κ-compact objects under colimits. By Theorem 5.5.1.1 of [Lur08], this is equivalent to the condition $\mathcal{C} = \mathrm{Ind}_\kappa(\mathcal{C}^\kappa)$.

We say $\mathcal{C}$ is presentable if it is κ-presentable for some κ. Again by Theorem 5.5.1.1 of [Lur08], this is equivalent to the condition that $\mathcal{C}$ is a localization of a presheaf category.

Definition 3.1.1.2. We define $\mathbf{Pr}^L$ to be the non-locally small $(\infty,1)$-category whose objects are the class of the presentable categories and morphisms the colimit preserving functors (technically speaking, we cannot define $\mathbf{Pr}^L$ because its morphisms form a class). We define $\mathbf{Pr}^L_\kappa \subset \mathbf{Pr}^L$ to be the 1-full subcategory with objects the κ-presentable categories and morphisms the functors preserving κ-compact objects. This time it is indeed locally small.

Theorem 3.1.1.3. We have $\mathbf{Pr}^L_\kappa \in \mathbf{Pr}^L_\kappa$.

Proof: By the equivalent incarnation of κ-presentable categories, we have $\mathbf{Pr}^L_\kappa$ equivalent to the categories that are of the form $\mathcal{C}^\kappa$ for $\mathcal{C} \in \mathbf{Pr}^L_\kappa$. This condition is formulated as $\mathcal{C}^\kappa$ is small, admits all κ-small colimits and retracts, and the functors between them preserving κ-small colimits. We denote this equivalent category by Rex_κ. We now prove Rex_κ is κ-presentable.

Note that the forgetful functor $U : \mathrm{Rex}_\kappa \to \mathbf{Cat}$ admit a left adjoint given by $\mathcal{C} \mapsto \mathrm{pSh}(\mathcal{C})^\kappa$. We claim that this is a pair of monadic adjunction, so that Rex_κ is the algebra of the monad $U \circ F$. By Proposition 4.7.3.5 of [Lur17], it suffices to prove that U is conservative, Rex_κ admits and U preserves U-split geometric realizations. Clearly U is conservative (it is faithful). Then, let $\mathcal{C}_\bullet$ be a simplicial diagram in Rex_κ, admitting a geometric realization $\mathcal{C}_{-1} \in \mathbf{Cat}$, with a splitting of $\mathcal{C}_\bullet \to \mathcal{C}_{-1} \in \mathbf{Cat}$. But then, $\mathrm{Fun}(I,\mathcal{C}_\bullet) \to \mathrm{Fun}(I,\mathcal{C}_{-1})$ is a split geometric realization in $\mathbf{Cat}$ for any κ-small diagram I, extending to $I^\triangleright$ shows that $\mathcal{C}_{-1} \in \mathrm{Rex}_\kappa$.

Finally, we use this to show that Rex_κ is κ-presentable. Firstly it admit all colimits because it is the algebra of a monad, and $\mathbf{Cat}$ admits all colimits. Note that $\mathbf{Cat}$ is the category of all small categories, hence compactly generated by the categories Δ^n. Upon showing U preserves κ-filtered colimits, we can show F preserves κ-compact objects, which shows that Rex_κ is κ-compactly generated. For a κ-filtered diagram $\mathcal{D}_i \in \mathrm{Rex}_\kappa$, their κ-filtered colimit in $\mathbf{Cat}$ $\mathcal{D}$ admits all κ-small colimits by the construction of colimit categories, and the functors $\mathcal{D}_i \to \mathcal{D}$ preserves κ-small colimits. This finishes the proof. □

We recall more facts on the category $\mathbf{Pr}^L$.

Theorem 3.1.1.4 (Adjoint functor theorem, [Lur08] Corollary 5.5.2.9). Let $F : \mathcal{C} \to \mathcal{D}$ be a functor between presentable categories. Then
(1) F admits a right adjoint iff F preserves all colimits.
(2) F admits a left adjoint iff F preserves all limits and preserves κ-filtered colimits for some regular cardinal κ. In this case, its left adjoint preserves κ-compact objects.

Definition 3.1.1.5. We define $\mathbf{Pr}^R$ and $\mathbf{Pr}^R_\kappa$ to be the categories of (κ-) presentable categories with morphisms right adjoints. They are opposite to $\mathbf{Pr}^L$ and $\mathbf{Pr}^L_\kappa$ respectively.

Theorem 3.1.1.6 ([Lur17], Proposition 5.5.3.13, Theorem 5.5.3.18, Proposition 5.5.7.6). The category $\mathbf{Pr}^L$ admits all limits, and the forgetful functor $\mathbf{Pr}^L \to \widehat{\mathbf{Cat}}$ preserves limits.

The category $\mathbf{Pr}^L$ admits all colimits, equivalently, the category of $\mathbf{Pr}^R$ admits all limits, and the forgetful functor $\mathbf{Pr}^R \to \widehat{\mathbf{Cat}}$ preserves limits.

The analogous statements hold for $\mathbf{Pr}^L_\kappa$ and $\mathbf{Pr}^R_\kappa$.

Then, we recall the definition of Lurie tensor products of presentable categories.

Construction 3.1.1.7. For $\mathcal{C}, \mathcal{D} \in \mathbf{Pr}^L$, there is a category $\mathcal{C} \otimes \mathcal{D} \in \mathbf{Pr}^L$, universal among presentable categories with a functor $\mathcal{C} \times \mathcal{D} \to \mathcal{C} \otimes \mathcal{D}$. An analogous construction works for $\mathcal{C}, \mathcal{D} \in \mathbf{Pr}^L_\kappa$.

The construction goes as follows: suppose $\mathcal{C} = \mathrm{pSh}(\mathcal{C}_0)[W^{-1}]$, and $\mathcal{D} = \mathrm{pSh}(\mathcal{D}_0)[V^{-1}]$, we have

$$\mathcal{C} \otimes \mathcal{D} = \mathrm{pSh}(\mathcal{C}_0 \times \mathcal{D}_0)[W^{-1} \times V^{-1}]$$

It can be verified that this gives a symmetric monoidal structure on $\mathbf{Pr}^L$ and $\mathbf{Pr}^L_\kappa$.

Theorem 3.1.1.8. The category $\mathbf{Pr}^L$ is Cartesian closed. The internal Hom object is given by $\mathrm{Fun}^L(-,-)$. Similarly, the category of $\mathbf{Pr}^L_\kappa$ has internal Hom object $\mathrm{Ind}_\kappa \mathrm{Fun}_{\mathrm{Rex}_\kappa}((-)^\kappa, (-)^\kappa)$.

3.1.2. Stefanich rings and higher modules

Definition 3.1.2.1. Recall from Theorem 3.1.1.3 that we have $\mathbf{Pr}^L_\kappa \in \mathbf{Pr}^L_\kappa$. We iteratively define $(n+1)\mathbf{Pr}^L_\kappa := \mathbf{Mod}_{n\mathbf{Pr}^L_\kappa}(\mathbf{Pr}^L_\kappa)$, where $1\mathbf{Pr}^L_\kappa := \mathbf{Pr}^L_\kappa$, $0\mathbf{Pr}^L_\kappa = \mathbf{Ani}$.

This should be think of the analogous fact that $(\infty, 2)$-categories are in some sense categories enriched in $(\infty, 1)$-categories.

Theorem 3.1.2.2. Let $\mathcal{C} \in \mathbf{CAlg}(\mathbf{Pr}^L_\kappa)$ be a presentably symmetric monoidal category. Then, the functor $\mathbf{CAlg}(\mathcal{C}) \to \mathbf{CAlg}(\mathbf{Mod}_{\mathcal{C}}(\mathbf{Pr}^L_\kappa))$ given by $A \mapsto \mathbf{Mod}_A$ (or $\mathbf{LMod}_A$ equivalently, if we want

the explicit $\mathbb{E}_\infty$-tensor structure on the category) is fully faithful, and admits a right adjoint given by $\mathcal{D} \mapsto \mathrm{End}_{\mathcal{D}}(\mathbb{1}_{\mathcal{D}})$. In particular, $\mathrm{End}_{\mathbf{Mod}_A}(A) = A$.

remark 3.1.2.1. The formula $\mathbf{Mod}_A \otimes_{\mathbf{Mod}_B} \mathbf{Mod}_C = \mathbf{Mod}_{A \otimes_B C}$ can be regarded as the commutation of the above functor with pushouts.

From now on, we often fix $\kappa = \aleph_1$ (the cardinality $2^{\aleph_0}$), and omit κ's and L's from all $\mathbf{Pr}^L_\kappa$'s when we do so.

Definition 3.1.2.3. We define the κ-presentably symmetric monoidal $(\infty, 1)$-category of Stefanich rings to be

$$\mathbf{StRing} = \operatorname*{colim}_{\mathbf{Mod}_{(-)}} (\mathbf{CAlg}(0\mathbf{Pr}) \hookrightarrow \mathbf{CAlg}(1\mathbf{Pr}) \hookrightarrow \mathbf{CAlg}(2\mathbf{Pr}) \hookrightarrow \cdots)$$

where the colimit is taken in $\mathbf{Pr}^L_\kappa$, the transition map given as in Theorem 3.1.2.2. Equivalently, we can define

$$\mathbf{StRing} = \lim_{\mathrm{End}_{(-)}} (\mathbf{CAlg}(0\mathbf{Pr}) \leftarrow \mathbf{CAlg}(1\mathbf{Pr}) \leftarrow \mathbf{CAlg}(2\mathbf{Pr}) \leftarrow \cdots)$$

this time computing the limit as plain categories.

Lemma 3.1.2.4. Let $(\mathcal{C}_n)_{n \geqslant n_0}$ be a sequence of objects of $\mathbf{Pr}^L_\kappa$, with transition functors fully faithful left adjoints $F_n : \mathcal{C}_n \hookrightarrow \mathcal{C}_{n+1}$. Then, the inclusion

$$\iota_n : \mathcal{C}_n \hookrightarrow \mathcal{C} := \operatorname*{colim}_{\mathbf{Pr}^L_\kappa} \left(\mathcal{C}_{n_0} \hookrightarrow \mathcal{C}_{n_0+1} \hookrightarrow \cdots \right)$$

is fully faithful.

Proof: We denote the right adjoint to F_n by G_n. Since $\{m | m \geqslant n\} \subset \{m | m \geqslant n_0\}$ is cofinal, we may assume $n = n_0$. Then,

$$\mathcal{C} = \lim_{\mathbf{Cat}} (\cdots \to \mathcal{C}_{n+1} \to \mathcal{C}_n)$$

Then, the above inclusion is left adjoint to the projection $q_n : \mathcal{C} \twoheadrightarrow \mathcal{C}_n$. We must show that the unit map $\mathrm{id}_{\mathcal{C}_n} \to q_n \iota_n$ is an equivalence. Let $c \in \mathcal{C}_n$, we claim that $\iota_n(c)$ is computed by the sequence of objects $(c, F_n(c), F_{n+1} \circ F_n(c), \cdots)$. To prove this, take any $(d_n, d_{n+1}, \cdots) \in \mathcal{C}$, consider

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}}((c, F_n(c), \cdots), (d_n, d_{n+1}, \cdots)) &= \lim_{j \geqslant n} \mathrm{Hom}_{\mathcal{C}_j}(F_j \circ \cdots \circ F_n(c), d_j) \\ &= \lim_{j \geqslant n} \mathrm{Hom}_{\mathcal{C}_n}(c, G_n \circ \cdots \circ G_j(d_j)) \\ &= \mathrm{Hom}_{\mathcal{C}_n}(c, d_n) \end{aligned}$$

Thus, the composition is obviously the identity. □

We now fix an appropriate category of commutative algebras $\mathbf{CAlg}$, for example, $\mathbf{CAlg} = \mathbf{CAlg}(\mathbf{Sp})$.

Construction 3.1.2.5. By the above lemma, the faithful inclusion $\mathbf{Mod} : \mathbf{CAlg} \hookrightarrow \mathbf{CAlg}(1\mathbf{Pr})$ gives rise to a fully faithful functor

$$\mathcal{O} : \mathbf{CAlg} \hookrightarrow \mathbf{CAlg}(1\mathbf{Pr}) \hookrightarrow \mathbf{StRing}$$

We take $\mathbf{Aff} = \mathbf{CAlg}^{\mathrm{op}}$, and also denote the functor

$$\mathcal{O} : \mathbf{Aff}^{\mathrm{op}} \hookrightarrow \mathbf{StRing}$$

Take $\operatorname{Spec} f : \operatorname{Spec} B \to \operatorname{Spec} A \in \mathbf{Aff}$. The n-th component of $\mathcal{O}(\operatorname{Spec} f) : \mathcal{O}(\operatorname{Spec} A) \to \mathcal{O}(\operatorname{Spec} B)$ is given by

$$f_n^* : n\mathbf{Pr}_A \to n\mathbf{Pr}_B$$

Explicitly, $f_0^* : \mathbf{Mod}_A \to \mathbf{Mod}_B$ is given by $- \otimes_A B$, and

$$f_n^* : \mathbf{Mod}_{(n-1)\mathbf{Pr}_A}(1\mathbf{Pr}) \to \mathbf{Mod}_{(n-1)\mathbf{Pr}_B}(1\mathbf{Pr})$$

is given by $- \otimes_{(n-1)\mathbf{Pr}_A} (n-1)\mathbf{Pr}_B$, with the structure morphism given by f_{n-1}^*.

We will then show that the above data of f_n^* can be upgraded to a six-functor formalism over $\mathbf{Aff}$.

Definition 3.1.2.6. Let $\mathcal{C} \in \mathbf{CAlg}(\mathbf{Pr}_L^\kappa)$, and $f : A \to B \in \mathbf{CAlg}(\mathcal{C})$. We say f is κ-finitely presented if $B \in \mathbf{CAlg}(\mathbf{Mod}_A(\mathcal{C}))^\kappa$.

Lemma 3.1.2.7. Let $\mathcal{C} \in \mathbf{CAlg}(\mathbf{Pr}_L)$. Then, the forgetful functor $\mathbf{CAlg}(\mathcal{C}) \to \mathcal{C}$ preserves sifted colimits.

Proof: We denote $F : \mathcal{C} \rightleftarrows \mathbf{CAlg}(\mathcal{C}) : U$ the free-forgetful adjunction. Let I be any sifted diagram. Then, take $i \mapsto A_i \in \mathbf{CAlg}(\mathcal{C})$, and let $M = \operatorname{colim}_I UA_i$. Then, we construct an $\mathbb{E}_\infty$-algebra structure on M such that it becomes equivalent to $\operatorname{colim}_I A_i$. The structural maps are given by the identifications

$$M^{\otimes n} = \operatorname*{colim}_{I^n} UA_{i_1} \otimes \cdots UA_{i_n} = \operatorname*{colim}_I A_i^{\otimes n}$$

where the first equivalence is due to the distribution law on colimits and tensors, and the second by the assumption that I is sifted. The above identification gives an $\mathbb{E}_\infty$-structure on M, which we will denote the corresponding algebra by $\widetilde{M}$. Equivalently, expressing $\widetilde{M}$ by the monad UF, it is given by

$$TM = \operatorname{colim} TUA_i \to \operatorname{colim} UA_i = M$$

Then, for any algebra A, since the adjunction $F \dashv U$ is monadic, we have an equivalence

$$A = \operatorname*{colim}_{\Delta^{\mathrm{op}}} F(UF)^n UA$$

Hence, we have the chain of equivalences

$$\begin{aligned}\mathrm{Hom}_{\mathbf{CAlg}(\mathcal{C})}\big(\widetilde{M},B\big) &= \lim_{\Delta}\mathrm{Hom}_{\mathcal{C}}\big((UF)^nU\widetilde{M},UB\big)\\ &= \lim_{\Delta}\mathrm{Hom}_{\mathcal{C}}((UF)^nM,UB)\\ &= \lim_{\Delta}\lim_{I^{\mathrm{op}}}\mathrm{Hom}_{\mathcal{C}}((UF)^nUA_i,UB)\\ &= \lim_{I^{\mathrm{op}}}\lim_{\Delta}\mathrm{Hom}_{\mathcal{C}}(F(UF)^nUA_i,B)\\ &= \lim_{I^{\mathrm{op}}}\mathrm{Hom}_{\mathcal{C}}(A_i,B)\end{aligned}$$

hence expressing $\widetilde{M}=\mathrm{colim}_I\, A_i$. □

Lemma 3.1.2.8. Let $\mathcal{C}\in\mathbf{CAlg}(\mathbf{Pr}^\kappa_L)$. Then the functor $\mathbf{Mod}_{(-)}:\mathbf{CAlg}(\mathcal{C})\to\mathbf{CAlg}(\mathbf{Mod}_{\mathcal{C}}(\mathbf{Pr}^\kappa_L))$ is a functor in $\mathbf{Pr}^\kappa_L$. Namely, we have $\mathbf{Mod}_{(-)}$ preserves κ-compact objects and equivalently its right adjoint $\mathrm{End}_{(-)}(\mathbb{1})$ preserves κ-filtered colimits.

Proof: We prove the statement on the right adjoint $\mathrm{End}_{(-)}(\mathbb{1})$. Firstly, in the proof of Theorem 3.1.1.3, we have shown that the forgetful functor $\mathrm{Rex}_\kappa\to\mathbf{Cat}$ preserves κ-filtered colimits. Now, take I an κ-filtered diagram, $\mathcal{D}_i\in\mathbf{CAlg}(\mathbf{Mod}_{\mathcal{C}}(\mathbf{Pr}^\kappa_L))$, $\mathcal{D}=\mathrm{colim}_I\,\mathcal{D}_i$, and algebras $A_i=\mathrm{End}_{\mathcal{D}_i}(\mathbb{1}_i)$, $A=\mathrm{End}_{\mathcal{D}}(\mathbb{1})$. We must show the colimit taken in $\mathbf{CAlg}(\mathcal{C})$ $\mathrm{colim}_I\,A_i\to A$ is equivalent to A.

By the claim that $\mathrm{Rex}_\kappa\to\mathbf{Cat}$ preserves κ-filtered colimits, $\mathcal{D}=\mathrm{colim}_I\,\mathcal{D}_i$ gives $\mathcal{D}^\kappa=\mathrm{colim}_I\,\mathcal{D}_i^\kappa$, where the right colimit is taken in $\mathbf{Cat}$. It suffices to show that $\mathrm{Hom}(c,\mathrm{colim}\,A_i)=\mathrm{Hom}(c,A)$ for all $c\in\mathcal{C}^\kappa$.

By compactness of c, we have

$$\mathrm{Hom}_{\mathcal{C}}(c,\mathrm{colim}\,A_i)=\mathrm{colim}\,\mathrm{Hom}_{\mathcal{C}}(c,A_i)$$

Then, the algebra units $\mathcal{C}\to\mathcal{D}_i$ given by $c\mapsto c\otimes\mathbb{1}_i$ is a morhpism in $\mathbf{Pr}^L_\kappa$, so preserves κ-compact objects. Moreover, it admit a right adjoint given by $\mathrm{Hom}(\mathbb{1}_i,-)$. Thus, we have equivalences

$$\begin{aligned}\mathrm{colim}\,\mathrm{Hom}_{\mathcal{C}}(c,A_i) &= \mathrm{colim}\,\mathrm{Hom}_{\mathcal{D}_i}(c\otimes\mathbb{1}_i,\mathbb{1}_i)\\ &= \mathrm{Hom}_{\mathcal{D}}(c\otimes\mathbb{1},\mathbb{1})=\mathrm{Hom}_{\mathcal{C}}(c,A)\end{aligned}$$

□

Theorem 3.1.2.9. Let $\mathcal{C}\in\mathbf{CAlg}(\mathbf{Pr}^\kappa_L)$. Let $f:A\to B$ be a κ-finitely presented map. Then, the induced map $g^*:\mathbf{Mod}_A\to\mathbf{Mod}_B$ is also κ-finitely presented.

Proof: Apply Lemma 3.1.2.8 to $\mathcal{C}=\mathbf{Mod}_A(\mathcal{C})$. □

Corollary 3.1.2.10. Let $f:A\to B$ be a κ-finitely presented map. Then for each $n\geqslant 0$, by iteratively applying the previous theorem, we have $f^*_n:n\mathbf{Pr}_A\to n\mathbf{Pr}_B$ is a κ-finitely presented functor.

Lemma 3.1.2.11. Let $\mathcal{C} \in \mathbf{CAlg}(\mathbf{Pr}^L_\kappa)$. Then for $A \in \mathbf{CAlg}(\mathcal{C})$, we have A is κ-compact iff its underlying object is κ-compact.

Proof: Consider the free-forgetful adjunction $F : \mathcal{C} \rightleftarrows \mathbf{CAlg}(\mathcal{C}) : U$. Note that for $M \in \mathcal{C}^\kappa$, we have

$$U \circ F(M) = \bigoplus_{n \in \mathbb{N}} (M^{\otimes n})_{h\Sigma_n}$$

which is a countable direct sum of κ-compact objects, hence κ-compact since κ is uncountable. Then, by Lemma 3.1.2.7, U preserves sifted colimits hence κ-filtered colimits, so F preserves κ-compact objects, hence $F(M)$ is compact in $\mathbf{CAlg}(\mathcal{C})$. Also, notice that $F(M)$ generate the category $\mathbf{CAlg}(\mathcal{C})^\kappa$ under retracts and κ-small colimits. So, now it suffices to prove that algebras with underlying modules κ-compact are closed under retracts and κ-small colimits.

For retracts the claim is obvious. For κ-small colimits, we represent the colimit using its simplicial replacement. This refers to the following general procedure: let I be any diagram and A_i are objects indexed by this diagram, we have

$$\operatorname*{colim}_I A_i = \operatorname*{colim}_{[n] \in \Delta^{\mathrm{op}}} \coprod_{f \in \operatorname{Hom}(\Delta^n, I)} f(0)$$

with the obvious transition maps. For I a κ-small diagram, each coproduct in the simplicial replacement is still κ-small. Then, by Lemma 3.1.2.7, the functor U preserves geometric realizations, so it suffices to prove $U(\coprod f(0))$ is κ-small, which is by assumption. Thus, we have proven that every compact algebra is compact as a module.

For the converse, suppose $U(A) \in \mathcal{C}$ is compact. Then by monadicity we have $A = \operatorname{colim} (FU)^n A$. However every term in the simplicial diagram is κ-compact as an algebra, we obtain that A is also κ-compact. □

Lemma 3.1.2.12. Let $\mathcal{C} \in \mathbf{CAlg}(\mathbf{Pr}^L_\kappa)$. Then, for $A \in \mathbf{CAlg}(\mathcal{C})$, if A is a κ-finitely presented algebra, then $M \in \mathbf{Mod}_A(\mathcal{C})$ is κ-compact iff its underlying object is κ-compact.

Proof: Let $M \in \mathbf{Mod}_A(\mathcal{C})^\kappa$. Note the κ-compact A-modules are generated by free A-modules generated by κ-compact objects under retracts and κ-small colimits. Hence, since the forgetful functor preserves colimits, it suffices to prove that free A-modules generated by κ-compact objects have their underlying objects compact. This is obvious, since A is κ-compact and hence $A^{\oplus \alpha}$ is κ-compact for all $\alpha < \kappa$.

Then, suppose the underlying object of M is compact. The argument follows again from the monadic free-forgetful adjunction as in the previous proof of Lemma 3.1.2.11. □

Then, we show that the above data can be assembled into a six-functor formalism (see Section 3.3.1).

Theorem 3.1.2.13. For every $n \geqslant 0$, we have a six-functor formalism $n\mathbf{Pr}^{*,*}$ given by

$$\operatorname{Span}(\mathbf{Aff}, \text{all}, \aleph_1\text{-finitely presented maps}) \to (n+1)\mathbf{Pr}$$

$$\operatorname{Spec} A \mapsto n\mathbf{Pr}_A, \left(\operatorname{Spec} B \xleftarrow{f} \operatorname{Spec} A \xrightarrow{g} \operatorname{Spec} C\right) \longmapsto \left(n\mathbf{Pr}_B \xrightarrow{f_n^*} n\mathbf{Pr}_A \xrightarrow{g_{n,*}} n\mathbf{Pr}_C\right)$$

Explicitly, we make the following requirements:

(1) $n\mathbf{Pr}_A$ is a Cartesian closed symmetric monoidal $\aleph_1$-presentable $(\infty, n+1)$-category.

(2) For all maps $\operatorname{Spec} f : \operatorname{Spec} A \to \operatorname{Spec} B$, there is a pullback functor $f_n^* : n\mathbf{Pr}_B \to n\mathbf{Pr}_A$, preserving all colimits and canonically compatible with composition of functors.

(3) For $\aleph_1$-finitely presented maps $\operatorname{Spec} g : \operatorname{Spec} A \to \operatorname{Spec} C$ (i.e., the map $g : C \to A$ is $\aleph_1$-finitely presented), there is a pushforward functor $g_{n,*}$, preserving all colimits and canonically compatible with composition of functors.

(4) For any Cartesian diagram in **Aff**

$$\begin{array}{ccc} \operatorname{Spec} A' & \xrightarrow{g'} & \operatorname{Spec} A \\ {\scriptstyle f'}\downarrow & \ulcorner & \downarrow{\scriptstyle f} \\ \operatorname{Spec} B' & \xrightarrow[g]{} & \operatorname{Spec} B \end{array}$$

with f hence f' $\aleph_1$-finitely presented, we have a canonical base change isomorphism

$$g_n^* \circ f_{n,*} = f'_{n,*} \circ g_n'^*$$

compatible with horizontal and vertical composition of Cartesian squares.

(5) For each $f : \operatorname{Spec} B \to \operatorname{Spec} A$ $\aleph_1$-finitely presented, a canonical projection formula isomorphism

$$f_{n,*}M \otimes N = f_{n,*}(M \otimes f_n^* N)$$

for any $M \in n\mathbf{Pr}_B, N \in n\mathbf{Pr}_A$

Proof: We define $g_{n,*}$ to be the right adjoint of g_n^*, the forgetful functor or restriction of scalars. It naturally admits a right adjoint given by coinduction, so we verify that it preserves $\aleph_1$-compact objects. For $g : C \to A$ $\aleph_1$-finitely presented, by Corollary 3.1.2.10 we have $g_m^* : m\mathbf{Pr}_C \to m\mathbf{Pr}_A$ $\aleph_1$-finitely presented for all m. Then, by applying Lemma 3.1.2.12 with $\mathcal{C} = n\mathbf{Pr}_A$, $A = (n-1)\mathbf{Pr}_B$, we have $g_{n,*}$ preserves $\aleph_1$-compact object, hence defines a morphism in $\mathbf{Pr}^L_{\aleph_1}$.

Now, the functorialities are all obvious, and we now unwind the required conditions of the base change isomorphism and projection formula.

For the base change isomorphism, consider a pushout diagram of algebras $D = B \otimes_A C$. Then, the category of modules fit into the following square

$$\begin{array}{ccc} \mathbf{Mod}_A & \xrightarrow{f'^*} & \mathbf{Mod}_B \\ {\scriptstyle g'_*}\uparrow & & \uparrow{\scriptstyle g_*} \\ \mathbf{Mod}_C & \xrightarrow[f^*]{} & \mathbf{Mod}_D \end{array}$$

since unwinding definitions, let $M \in \mathbf{Mod}_C$, $f'^* \circ g'_*(M) = B \otimes_A M$, $g_* \circ f^*(M) = D \otimes_C M$, and we have

$$D \otimes_C M = (B \otimes_A C) \otimes_C M = B \otimes_A M$$

the base change isomorphism. Then, take A, B, C, D to be the corrsponding $(n-1)\mathbf{Pr}$'s give the original base change isomorphism.

Now it suffices to treat the projection formula. Take $f : A \to B$ a map of algebras, and take modules $M \in \mathbf{Mod}_B$, $N \in \mathbf{Mod}_A$. Then, the projection formula reads

$$M \otimes_B f_*(N) = f_*(f^*(M) \otimes_A N)$$

Note that both sides preserves colimits in M, so we may assume that M is free, and in which case the assertion is obvious. Finally taking A and B to be the corrsponding $(n-1)\mathbf{Pr}$'s give the original projection formula result. □

remark 3.1.2.2. The requirements that g's should be $\aleph_1$-finitely presented could be removed, and the above construction will have result a six-functor formalism in $\mathbf{Pr}^L$ instead of $\mathbf{Pr}^L_\kappa$.

3.1.3. Further properties of the six-functor formalism

Proposition 3.1.3.1. Let $\mathcal{C}$ be an $(\infty, 3)$-category. Let $F : c \to d$ be a morphism in $\mathcal{C}$, then the following ara equivalent.

(1) F admits a right adjoint $F^R : d \to c$, and both $\eta : \mathrm{id}_c \to F^R \circ F \in \mathrm{End}(c)$ and $\varepsilon : F \circ F^R \to \mathrm{id}_d \in \mathrm{End}(d)$ admit right adjoints, in the sense that $\mathrm{End}(c)$ and $\mathrm{End}(d)$ are $(\infty, 2)$-categories.

(2) F admits a left adjoint $F^L : d \to c$, and both $\eta : \mathrm{id}_d : F \circ F^L \in \mathrm{End}(d)$ and $\varepsilon : F^R \circ F \to \mathrm{id}_c$ admit left adjoints.

When the equivalent condition holds, we have a natural identification $F^L \simeq F^R$. One also call this pair of morphisms by fully adjointable morphisms.

Proof: We prove that under the condition of (1), we have F^R also the left adjoint of F. The other direction is dual.

We will attempt to find $\eta' : \mathrm{id}_d \to F \circ F^R \in \mathrm{End}(d)$ and $\varepsilon' : F^R \circ F \to \mathrm{id}_c \in \mathrm{End}(c)$ such that they are the data of unit and counit for $F^R \dashv F$. We define $\eta' := \varepsilon^R$ and $\varepsilon' := \eta^R$. One can easily check that these data satisfies the condition (2). □

Corollary 3.1.3.2. Let $\operatorname{Spec} f : \operatorname{Spec} B \to \operatorname{Spec} A$ be a morphism in **Aff**. Suppose $f : A \to B$ is $\aleph_1$-finitely presented, then for $n \geqslant 1$, we have the functor $f_n^* : n\mathbf{Pr}_A \to n\mathbf{Pr}_B$ satisfy the equivalent conditions of Proposition 3.1.3.1. In particular, $f_{n,*} : n\mathbf{Pr}_B \to n\mathbf{Pr}_A$ is also left adjoint to f_n^*.

Proof: Firstly note that we may take the ambient category to be $(n+1)\mathbf{Pr}_A$, which lies in $(n+2)\mathbf{Pr}$, so admits an $(\infty, 3)$-category structure. Now, it suffices to prove that the unit and counit of the adjunction $f_n^* \dashv f_{n,*}$ admit right adjoints.

For the unit $\mathrm{id}_{n\mathbf{Pr}_A} \to f_{n,*}f_n^*$, its right adjoint would be a natural transformation $f_{n,*}f_n^* \to \mathrm{id}_{n\mathbf{Pr}_A}$, such that there are further units and counits such that the adjunction identity holds. But note here that everything is $n\mathbf{Pr}_A$-linear, it suffices to construct the right adjoint of $\left(\mathrm{id}_{n\mathbf{Pr}_A} \to f_{n,*}f_n^*\right)((n-1)\mathbf{Pr}_A)$, and extend linearly to a full right adjoint. Here, we can identify the functor simply as $f_{n-1}^* : (n-1)\mathbf{Pr}_A \to (n-1)\mathbf{Pr}_B$, which admit an obvious right adjoint.

For the counit $f_n^* f_{n,*} \to \mathrm{id}_{n\mathbf{Pr}_B}$, we consider the following identification:

$$\begin{aligned}\mathrm{End}_{(n+1)\mathbf{Pr}_A}(n\mathbf{Pr}_B) &= \mathrm{End}_{\mathbf{Mod}_{n\mathbf{Pr}_A}(1\mathbf{Pr})}\left(\mathbf{Mod}_{(n-1)\mathbf{Pr}_B}(n\mathbf{Pr}_A)\right)\\ &= \mathbf{Mod}_{(n-1)\mathbf{Pr}_B}(n\mathbf{Pr}_A) \otimes_{\mathbf{Mod}_{(n-1)\mathbf{Pr}_A}(n\mathbf{Pr}_A)} \mathbf{Mod}_{(n-1)\mathbf{Pr}_B}(n\mathbf{Pr}_A)\\ &\quad \text{(since module categories are self-dual)}\\ &= \mathbf{Mod}_{(n-1)\mathbf{Pr}_B \otimes_{(n-1)\mathbf{Pr}_A}(n-1)\mathbf{Pr}_B}(n\mathbf{Pr}_A)\end{aligned}$$

Under this identification, the counit is simply the diagonal

$$\Delta_{n-1}^* : (n-1)\mathbf{Pr}_B \otimes_{(n-1)\mathbf{Pr}_A} (n-1)\mathbf{Pr}_B \to (n-1)\mathbf{Pr}_B$$

which admits a right adjoint. □

remark 3.1.3.1. As in the previous case, if we remove the $\aleph_1$-finitely presented condition, we will still get an ambidexterity result, but in the category $\mathbf{Pr}^L$ instead of $\mathbf{Pr}_\kappa^L$.

We now temporarily set $\mathbf{CAlg} = \mathbf{CAlg}(\mathbf{Sp})$, and prove a descent result.

Theorem 3.1.3.3. For $n \geqslant 1$, the functor $n\mathbf{Pr}^{*,*} : \mathbf{Aff}^{\mathrm{op}} \to (n+1)\mathbf{Pr}$ is a sheaf in the flat topology.

Proof: By construction, we have $n\mathbf{Pr}^{*,*}$ sends coproducts to products. Hence, it suffices to prove that for a flat hypercover of $\mathbb{E}_\infty$-rings $A \to A_\bullet$, the map $n\mathbf{Pr}_A \to \lim_\Delta^* n\mathbf{Pr}_{A_\bullet}$ is an equivalence (on the level of rings, we have slightly unfortunately that the functor f_n^* is covariant).

Firstly, we assume that $A \to A_\bullet$ is an $\aleph_1$-finitely presented hypercover (i.e., the maps $f^i : A \to A_i$ are $\aleph_1$-finitely presented). We prove in this case that the functor

$$0\mathbf{Pr}_A \to \lim_\Delta{}^! \, 0\mathbf{Pr}_{A_\bullet} = \operatorname*{colim}_{\Delta^{\mathrm{op}}}{}_* \, 0\mathbf{Pr}_{A_\bullet}$$

is fully faithful and essentially surjective. We note here that the functor $0\mathbf{Pr}_A \to \lim^! 0\mathbf{Pr}_{A_\bullet}$ given by !-pullback admit a left adjoint given by $(M_n) \mapsto \mathrm{colim}\, f_*^n M_n$. Equivalently, we compute this limit in $\mathbf{Pr}_{\aleph_1}^R$.

For full faithfulness, take $M \in 0\mathbf{Pr}_A$, we must show that the counit map

$$\operatorname*{colim}_{\Delta^{\mathrm{op}}} f_*^i f^{i,!} M \to M$$

an equivalence. Notice that by definition, we have $f^{i,!}M = \mathrm{Hom}_A(A_i, M)$ regarded as an A_i-module. Thus, we have $f_*^i f^{i,!} M = \mathrm{Hom}_A(A_i, M)$ the underlying A-module. Then, we may compute the colimit as

$$\operatorname*{colim}_{\Delta^{\mathrm{op}}} f_*^i f^{i,!} M = \operatorname*{colim}_{m \geqslant 0} \operatorname*{colim}_{\Delta^{\mathrm{op}}_{\leqslant m}} \mathrm{Hom}_A(A_i, M) = \operatorname*{colim}_{m \geqslant 0} \mathrm{Hom}_A\left(\lim_{\Delta_{\leqslant m}} A_i, M\right)$$

where the final equivalence is true because $\lim_{\Delta_{\leqslant m}}$ is a finite colimit, hence preserved by the exact functor $\mathrm{Hom}_A(-, M)$. Note that

$$M = \operatorname*{colim}_{m \geqslant 0} M = \operatorname*{colim}_{m \geqslant 0} \mathrm{Hom}_A(A, M)$$

thus it suffices to identify the pro-objects $\underline{\lim}\, A$ and $\underline{\lim} \lim_{\Delta_{\leqslant m}} A_i$.

By flatness, we may assume that A and A_i's are connective by replacing by their $\tau_{\geqslant 0}$'s. Then, to show that $\underline{\lim} A$ and $\underline{\lim} \lim_{\Delta_{\leqslant m}} A_i$ are equivalent as pro-objects, we may equivalently prove that the cofiber of $\underline{\lim} A \to \underline{\lim} \lim_{\Delta_{\leqslant m}} A_i$ is zero. This cofiber is given by the pro-object $\underline{\lim} C_m$, where $C_m = \mathrm{cofib}\big(A \to \lim_{\Delta_{\leqslant m}} A_i\big)$. Now, we prove that the transition maps $C_{m+1} \to C_m$ are phantom, and by Theorem 4.1.1.1 we have $C_{m+2} \to C_m$ is null. Hence, we have the whole pro-object is trivial.

To prove that these maps are phantom, we claim that $C_m[m]$ is a flat A-module. By the derived version of Lazard's theorem ([Lur17] 7.2.2.15), we may take $C_m = \mathrm{colim}\, A_\alpha^{\oplus n_\alpha}[-m]$, where the colimit is filtered. Hence, for any perfect A-module M, the map $M \to C_{m+1}$ factors through some $A_\alpha^{\oplus n_\alpha}[-m-1]$. Hence it suffices to show that $A_\alpha^{\oplus n_\alpha}[-m-1] \to C_m$ is null, which is by connectivity of A.

Finally, we derive the desired flatness. Again by flatness, we may assume that A is discrete. This time, by cosimplicial Dold-Kan correspondence, we may assume the cosimplicial object A_i is represented by the negative flat chain complex $0 \to N_0 \to N_{-1} \to \cdots$, where $\lim_{\Delta_{\leqslant m}} A_i$ is given by

$$0 \to N_0 \to N_{-1} \to \cdots \to N_{-m}$$

Hence C_m is given by

$$0 \to A \to N_0 \to N_{-1} \to \cdots \to N_{-m}$$

which is concentrated at degree $-m$. Hence this proves shifted flatness in the case where A is discrete, hence proves the general case.

Then we prove essential surjectivity. Consider the commutative square

$$\begin{array}{ccc} A & \longrightarrow & \underline{\lim} A_i \\ \downarrow & & \downarrow \\ A_0 & \longrightarrow & A_0 \otimes_A \underline{\lim} A_i \end{array}$$

We wish to show that $0\mathbf{Pr}_A = \lim^! 0\mathbf{Pr}_{A_i}$. To show this, it suffices to show the following three equivalences:

$$0\mathbf{Pr}_A = \lim_{\Delta}{}^! 0\mathbf{Pr}_{A_0^{\otimes_A n+1}}, 0\mathbf{Pr}_{A_0} = \lim_{\Delta}{}^! 0\mathbf{Pr}_{A_0 \otimes_A A_i}, 0\mathbf{Pr}_{A_i} = \lim_{\Delta}{}^! 0\mathbf{Pr}_{A_0^{\otimes_A n+1} \otimes_A A_i}$$

the second equivalence is by the splitting of the cosimplicial object, and the first and third equivalence is the case of a Cech cover. Thus, it suffices to prove that the essential surjectivity for the case where $A \to A_\bullet$ is a Cech cover.

Now, assume $A_i = A_0^{\otimes_A i+1}$ is a Cech cover. We wish to prove that

$$\lim_{\Delta}{}^! 0\mathbf{Pr}_{A_i} = \operatorname*{colim}_{\Delta^{\mathrm{op}}{}_*} 0\mathbf{Pr}_{A_i} = 0\mathbf{Pr}_A$$

Here, the colimit is computed in $\mathbf{Mod}_{0\mathbf{Pr}_A}\left(\mathbf{Pr}^L_{\aleph_1}\right)$. Also, the comparison functor in question $\operatorname*{colim}_{\Delta^{\mathrm{op}}{}_*} 0\mathbf{Pr}_{A_i} \to 0\mathbf{Pr}_A$ is an equivalence after applying $- \otimes_{0\mathbf{Pr}_A} 0\mathbf{Pr}_{A_0}$ by the existence of a splitting of the simplicial object. Moreover, since $A \to A_\bullet$ is a Cech cover, the comparison functor is an equivalence after applying all $- \otimes_{0\mathbf{Pr}_A} 0\mathbf{Pr}_{A_i}$, so is an equivalence after applying

$$- \otimes_{0\mathbf{Pr}_A} \operatorname*{colim}_{\Delta^{\mathrm{op}}{}_*} 0\mathbf{Pr}_{A_i}$$

Moreover, by the previous step we know that there is a fully faithful functor, whose post composition with the comparison functor is the identity

$$0\mathbf{Pr}_A \hookrightarrow \operatorname*{colim}_{\Delta^{\mathrm{op}}}{}_* \, 0\mathbf{Pr}_{A_i} \to 0\mathbf{Pr}_A$$

Hence the comparison functor is an equivalence after applying $- \otimes_{0\mathbf{Pr}_A} 0\mathbf{Pr}_A$, which is the identity.

We apply the above result for !-descent along countably presented faithfully flat hypercovers to prove the desired $*$-descent for countably presented faithfully flat hypercovers.

Consider the adjunction (note this is not a bi-adjunction due to the existence of lim)

$$F_{n,*} : \overset{*}{\lim} n\mathbf{Pr}_{A_\bullet} \rightleftarrows n\mathbf{Pr}_A : F_n^*$$

Here, F_n^* is the naive pullback, and $F_{n,*} : (M_i) \mapsto \operatorname{colim}_* F_{n,*} M_i$. We first show that F_n^* is fully faithful, or equivalently the counit map $F_{n,*}F_n^*$ is the identity. Since this map is $n\mathbf{Pr}_A$-linear, it suffices to show this counit map is the identity on the monoidal unit, i.e., the map

$$F_{n,*}F_n^*(n-1)\mathbf{Pr}_A \to (n-1)\mathbf{Pr}_A$$

is an equivalence. We prove this fact via inducion. For $n = 1$, note that the left hand side is precisely $\operatorname{colim}_* 0\mathbf{Pr}_{A_\bullet}$, which is the base case which we have proven above. For $n > 1$, we must prove $\operatorname{colim}_*(n-1)\mathbf{Pr}_{A_\bullet} \to (n-1)\mathbf{Pr}_A$ is an equivalence. However, upon taking dual, the above map corresponds to a map $(n-1)\mathbf{Pr}_A \to \lim^*(n-1)\mathbf{Pr}_{A_\bullet}$. Now, it suffices to show that this map is the hyperdescent comparison map for $n-1$ (here we use this map is an equivalence, which will be shown below). But this follows from the linearity and the image of the monoidal unit.

Now, it suffices to show that F_n^* is essentially surjective. But this follows from the same argument as in the case where $n = 0$.

Finally, we prove the $*$-descent for the general case. By reducing to the static case and choosing a filtered colimit presentation, we can prove that any faithfully flat map $A \to B$ is a $\aleph_1$-filtered colimit of countably presented faithfully flat map $A \to B_i$. Now, take $A \to A_0$ a flat cover, and suppose it is the filtered colimit of $A \to A_0^i$. We first show the descent for this case. We have

$$\lim_\Delta n\mathbf{Pr}_{A_0^{\otimes_A(m+1)}} = \lim_\Delta \operatorname*{colim}_I n\mathbf{Pr}_{(A_0^i)^{\otimes_A(m+1)}} = \operatorname*{colim}_I \lim_\Delta n\mathbf{Pr}_{(A_0^i)^{\otimes_A(m+1)}} = \operatorname*{colim}_I n\mathbf{Pr}_A = n\mathbf{Pr}_A$$

where the commuting of limits and colimits is by the $\aleph_1$-finiteness of Δ. For general flat hypercovers $A \to A_\bullet$, we can find

$$A \to A_\bullet = \operatorname*{colim}_I (A \to A_\bullet^i)$$

where each $A \to A_\bullet^i$ is a countably presented flat hypercover, by treating the matching maps one by one. By the same commuting as above, we have

$$\lim_\Delta n\mathbf{Pr}_{A_\bullet} = n\mathbf{Pr}_A$$

□

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remark 3.1.3.2. We remark here that it is important that we have set $1\mathbf{Pr}_A = \mathbf{Mod}_{0\mathbf{Pr}_A}\left(\mathbf{Pr}^L_{\aleph_1}\right)$ instead of $\mathbf{Mod}_{0\mathbf{Pr}_A}(\mathbf{Pr}^L)$.

There is a further result on general flat stacks. But this is not our main focus, so we just present the result here.

Theorem 3.1.3.4.

3.2. Geometry and higher category theory

3.2.1. The category of Stefanich rings

We begin with an analogue of the familiar fact $\mathbf{CAlg}_A(\mathcal{C}) = \mathbf{CAlg}(\mathcal{C})_{/A}$.

Lemma 3.2.1.1. The initial object in **StRing** is given by

$$\mathbb{P} = (*, 0\mathbf{Pr}, 1\mathbf{Pr}, \cdots)$$

Proof: This is by the construction of limits in categories. □

Construction 3.2.1.2. Let A be a Stefanich ring. We now construct the category of internal Stefanich rings in A.

The first ingredient we will use is the generalized $\mathbf{Mod}_n^A \dashv \mathrm{End}_n^A$ adjunction. For any n, by definition we have $A_n = \mathrm{End}_{A_{n+1}}(1)$. Applying the adjunction

$$\mathbf{Mod} : \mathbf{CAlg}(n\mathbf{Pr}) \rightleftarrows \mathbf{CAlg}((n+1)\mathbf{Pr}) : \mathrm{End}$$

The above equivalence gives a map by adjunction $L_n^A : \mathbf{Mod}_{A_n}(n\mathbf{Pr}) \to A_{n+1}$. Explicitly, this is the functor $M \mapsto 1 \otimes M$, where the tensoring comes from the equivalence $A_n = \mathrm{End}_{A_{n+1}}(1)$.

This map admits a right adjoint, given by the internal Hom

$$U_n^A : N \mapsto \mathrm{Hom}_{A_{n+1}}(1, N) \in \mathbf{Mod}_{\mathrm{End}_{A_{n+1}}(1)} = \mathbf{Mod}_{A_n}$$

which admits a further right adjoint by the cotensoring. Hence, we have L_n^A and U_n^A functors in $(n+1)\mathbf{Pr}$.

We then define

$$\mathbf{Mod}_n^A = L_n^A \circ \mathbf{Mod} : \mathbf{CAlg}(A_n) \to \mathbf{CAlg}(A_{n+1}), X \mapsto 1_{A_{n+1}} \otimes \mathbf{Mod}_X(A_n) \in \mathbf{CAlg}(A_{n+1})$$

whose right adjoint is given by

$$\mathrm{End}_n^A = \mathrm{End} \circ U_n^A : \mathbf{CAlg}(A_{n+1}) \to \mathbf{CAlg}(A_n), Y \mapsto \mathrm{End}_{\mathrm{Hom}_{A_{n+1}}(1,Y)}(1)$$

Lemma 3.2.1.3. The map $\mathbf{Mod}_n^A : \mathbf{CAlg}(A_n) \to \mathbf{CAlg}(A_{n+1})$ is fully faithful.

Proof: Note all functors are taken in the category $1\mathbf{Pr} = \mathbf{Pr}^L_{\aleph_1}$. Hence, it suffices to show that the unit $\mathrm{id} \to \mathrm{End}_n^A \circ \mathbf{Mod}_n^A$ is the identity on $\aleph_1$-compact objects.

Now, take $R \in \mathbf{CAlg}(A_n)$ a countably presented algebra. By definition, we have

$$\mathrm{End}_n^A \circ \mathbf{Mod}_n^A(R) = \mathrm{End}_{U_n^A L_n^A(\mathbf{Mod}_R(A_m))}(1) = \mathrm{End}_{T(\mathbf{Mod}_R(A_m))}(1)$$

where we temporarily denote $T = U_n^A L_n^A : \mathbf{Mod}_{A_n} \to \mathbf{Mod}_{A_n}$, and note that by definition T is a functor in $(n+1)\mathbf{Pr}$. Also by definition, we have the unit map $\mathrm{id} \to T$ the identity on A_n.

By definition, we have T a lax monoidal functor preserving adjunctions. Hence, applying T to the adjunction $F : A_n \rightleftarrows \mathbf{Mod}_R(A_n) : U$, we obtain an adjunction

$$T(F) : T(A_n) \rightleftarrows T(\mathbf{Mod}_R(A_n)) : T(U)$$

By lax monoidalness of T, we have $T(F)$ a symmetric monoidal functor, so we have the unit object of $T(\mathbf{Mod}_R(A_n))$ given by $T(F)\big(1_{T(A_n)}\big)$. Combining the above facts, we have the following computation.

$$\begin{aligned}
\mathrm{End}_{T(\mathbf{Mod}_R(A_m))}(1) &= \mathrm{End}_{T(\mathbf{Mod}_R(A_m))}\Big(T(F)\big(1_{T(A_n)}\big)\Big) \\
&= \mathrm{Hom}_{T(A_n)}\Big(1_{T(A_n)}, T(U)T(F)\big(1_{T(A_n)}\big)\Big) \text{ by adjunction} \\
&= \mathrm{Hom}_{T(A_n)}\Big(1_{T(A_n)}, T(R\otimes -)\big(1_{T(A_n)}\big)\Big) \text{ by definition of } U \text{ and } F \\
&= \mathrm{Hom}_{A_n}\big(1_{A_n}, R\otimes 1_{A_n}\big) \text{ since } A_n = T(A_n) \\
&= R
\end{aligned}$$

□

By definition $\mathbf{Mod}_n^A = L_n^A \circ \mathbf{Mod}$, which has $\mathbf{Mod}$ and $\mathbf{Mod}_n^A$ fully faithful functors. Unfortunately, L_n^A is not always fully faithful, but since the unit map of L_n^A is at least an equivalence on the identity object, we have the following partial full faithfulness.

Lemma 3.2.1.4. The functor $L_n^A : \mathbf{Mod}_{A_n}(n\mathbf{Pr}) \to A_{n+1}$ is fully faithful on dualizable objects (and hence, fully faithful on $\aleph_1$-filtered colimits of dualizable objects).

Proof: Let $X, Y \in \mathbf{Mod}_{A_n}(n\mathbf{Pr})$ be dualizable. Since L_n^A is symmetric monoidal, it preserves dualizable objects. Hence, we have identifications

$$\begin{array}{ccc}
\mathrm{Hom}_{\mathbf{Mod}_{A_n}(n\mathbf{Pr})}(X\otimes Y^\vee, 1) & \xrightarrow{\simeq} & \mathrm{Hom}_{\mathbf{Mod}_{A_n}(n\mathbf{Pr})}(X, Y) \\
\downarrow & & \downarrow \\
\mathrm{Hom}_{A_{n+1}}\big(L_n^A(X\otimes Y^\vee), 1\big) & \xrightarrow{\simeq} & \mathrm{Hom}_{A_{n+1}}\big(L_n^A(X), L_n^A(Y)\big)
\end{array}$$

Hence, it suffices to prove that the left arrow is an equivalence. But we have

$$\mathrm{Hom}_{\mathbf{Mod}_{A_n}(n\mathbf{Pr})}(X\otimes Y^\vee, 1) = \mathrm{Hom}_{\mathbf{Mod}_{A_n}(n\mathbf{Pr})}\big(X\otimes Y^\vee, U_n^A(1)\big) = \mathrm{Hom}_{A_{n+1}}\big(L_n^A(X\otimes Y^\vee), 1\big)$$

where the first equivalence is because $U_n^A L_n^A(A_n) = A_n$. □

Definition 3.2.1.5. Let $A \in \mathbf{StRing}$. We define the category of Stefanich rings internal to A to be the colimit

$$\mathbf{StRing}(A) = \operatorname*{colim}_{\mathbf{1Pr}}\left(\mathbf{CAlg}(A_1) \xhookrightarrow{\mathbf{Mod}_1^A} \mathbf{CAlg}(A_2) \xhookrightarrow{\mathbf{Mod}_2^A} \cdots\right)$$

Equivalently, it is the limit

$$\mathbf{StRing}(A) = \lim\left(\mathbf{CAlg}(A_1) \xleftarrow{\mathrm{End}_A^1} \mathbf{CAlg}(A_2) \xleftarrow{\mathrm{End}_A^2} \cdots\right)$$

Construction 3.2.1.6. Let $f : A \to B$ be a morphism of Stefanich rings. By definition, this is the data $f_{i-1}^* : A_i \to B_i$ (here the index in f denotes the categorical level, i.e., A_i and B_i are $(i-1)$-categories), if we denote $A = (A_0, A_1, \cdots)$ and $B = (B_0, B_1, \cdots)$.

We define $(B/A)_{n-1} = f_{n-1,*}\big(1_{B_n}\big)) \in \mathbf{CAlg}(A_n)$, where $1_{B_n} \in \mathbf{CAlg}(B_n)$ is the unit.

Theorem 3.2.1.7. We have an equivalence of categories

$$\mathbf{StRing}_{A/} \simeq \mathbf{StRing}(A)$$

Given by sending $A \to B$ to $((B/A)_0, (B/A)_1, \cdots)$.

Proof: By definition, we have an equivalence

$$\mathbf{StRing}_{A/} = \lim\left(\mathbf{CAlg}(0\mathbf{Pr})_{A_0/} \xleftarrow{\mathrm{End}_1^{\mathbb{P}}} \mathbf{CAlg}(1\mathbf{Pr})_{A_1/} \xleftarrow{\mathrm{End}_2^{\mathbb{P}}} \cdots\right)$$

Then, using $\mathbf{CAlg}(n\mathbf{Pr})_{A_n} = \mathbf{CAlg}\big(\mathbf{Mod}_{A_n}(n\mathbf{Pr})\big)$, the transition $\mathrm{End}_n^{\mathbb{P}}$ factors as

$$\mathbf{CAlg}\big(\mathbf{Mod}_{A_{n+1}}((n+1)\mathbf{Pr})\big) \xrightarrow{\mathrm{End}} \mathbf{CAlg}(A_{n+1}) \xrightarrow{U_m^A} \mathbf{CAlg}\big(\mathbf{Mod}_{A_n}(n\mathbf{Pr})\big)$$

Hence, by coinitialness, the category $\mathbf{StRing}_{A/}$ can be expressed as

$$\begin{aligned}
\mathbf{StRing}_{A/} &= \lim\left(\mathbf{CAlg}(0\mathbf{Pr})_{A_0/} \xleftarrow{\mathrm{End}_1^{\mathbb{P}}} \mathbf{CAlg}(1\mathbf{Pr})_{A_1/} \xleftarrow{\mathrm{End}_2^{\mathbb{P}}} \cdots\right) \\
&= \lim\big(\mathbf{CAlg}\big(\mathbf{Mod}_{A_0}(0\mathbf{Pr})\big) \leftarrow \mathbf{CAlg}\big(\mathbf{Mod}_{A_1}(1\mathbf{Pr})\big) \leftarrow \cdots\big) \\
&= \lim\big(\mathbf{CAlg}\big(\mathbf{Mod}_{A_0}(0\mathbf{Pr})\big) \leftarrow \mathbf{CAlg}(A_1) \leftarrow \mathbf{CAlg}\big(\mathbf{Mod}_{A_1}(1\mathbf{Pr})\big) \leftarrow \mathbf{CAlg}(A_2) \leftarrow \cdots\big) \\
&= \lim\left(\mathbf{CAlg}(A_1) \xleftarrow{\mathrm{End}_1^A} \mathbf{CAlg}(A_2) \xleftarrow{\mathrm{End}_2^A} \cdots\right) \\
&= \mathbf{StRing}(A)
\end{aligned}$$

The identification of this equivalence is direct. □

Corollary 3.2.1.8. The functor $\mathbf{StRing}_{A/} \to \mathbf{CAlg}(A_{n+1})$, $B \mapsto (B/A)_n$ admits a fully faithful left adjoint $\mathbf{CAlg}(A_{n+1}) \hookrightarrow \mathbf{StRing}(A) = \mathbf{StRing}_{A/}$.

We then prove some categorical properties of $\mathbf{StRing}(A)$. As in the case for general limits, we have limits and $\aleph_1$-filtered colimits in $\mathbf{StRing}$ are computed pointwise, since the trasition functors End_n^A preserves them. Arbitrary colimits, on the other hand, is much more difficult to compute. We will introduce a concept of pre-Stefanich rings and a "spectrification" functor (Lemma 3.2.1.10 and

Lemma 3.2.1.11) to compute them, just like the case when we are computing colimits in the category **Sp**.

Definition 3.2.1.9. Let A be a Stefanich ring. We define the category of pre-Stefanch rings $\mathbf{pStRing}(A)$ to be a sequence of objects $(B_0, B_1, \cdots)$, such that $B_n \in \mathbf{CAlg}(A_{n+1})$, and with structure comparison maps $\alpha_n : B_n \to \mathrm{End}_{A_{n+1}}(B_{n+1})$ (which we do **NOT** require to be an equivalence).

More precisely, we define

$$\mathcal{P}_A = \prod_{n \geqslant 0} \mathbf{CAlg}_{A_{n+1}}$$

and define an endofunctor

$$\mathrm{End}^A_\bullet : \mathcal{P}_A \to \mathcal{P}_A, (B_n) \mapsto (B'_n), B'_n = \mathrm{End}^A_{n+1}(B_{n+1})$$

Now, we may define the category of pre-Stefanich rings as the lax equalizer

$$\mathbf{pStRing}(A) = \mathcal{P}_A \ \vec{\times}_{\mathcal{P}_A \times \mathcal{P}_A} \ \mathcal{P}_A^{\Delta_1}$$

where $\mathcal{P}_A \to (\mathcal{P}_A \times \mathcal{P}_A)$ is given by $\left(\mathrm{id}, \mathrm{End}^A_\bullet\right)$. This is simply the categorical translation of the above definition.

Lemma 3.2.1.10. The category $\mathbf{pStRing}(A)$ admits all colimits. The colimits in $\mathbf{pStRing}(A)$ are computed pointwise.

Proof: Let $I \to \mathbf{pStRing}(A)$ be a diagram, with $i \mapsto (B^i, \alpha^i)$. Take

$$B = (\mathrm{colim}\, B^i) \in \mathcal{P}_A$$

which is the pointwise colimit. Then, we construct the structure map $B \to \mathrm{End}^A_\bullet(B)$. This is obtained by the lax commutation of $\mathrm{End}^A_\bullet$ with the colimit, explicitly

$$\alpha : B = \operatorname*{colim}_I B^i \to \operatorname*{colim}_I \mathrm{End}^A_\bullet(B^i) \to \mathrm{End}^A_\bullet\left(\operatorname*{colim}_I B^i\right) \to \mathrm{End}^A_\bullet(B)$$

where the second arrow is by the universal property of the colimit. Thus, we have $(B, \alpha) \in \mathbf{pStRing}(A)$. Finally we verify the universal property. By the construction of the lax pullback, for $(B, \alpha), (C, \beta) \in \mathbf{pStRing}(A)$, the Hom space is computed via

$$\mathrm{Hom}_{\mathbf{pStRing}(A)}((B, \alpha), (C, \beta)) = \mathrm{eq}\big(\mathrm{Hom}(B, Y) \rightrightarrows \mathrm{Hom}\big(B, \mathrm{End}^A_\bullet(Y)\big)\big)$$

where the two arrows are defined by $f \mapsto \beta \circ f$ and $f \mapsto \mathrm{End}^A_\bullet(f) \circ \alpha$. This verifies the universal property of the colimit by definition. □

Lemma 3.2.1.11. There exists a spectrification functor $L : \mathbf{pStRing}(A) \to \mathbf{StRing}(A)$, left adjoint to the inclusion $\mathbf{StRing}(A) \hookrightarrow \mathbf{pStRing}(A)$.

Proof: We define a functor $T : \mathbf{pStRing}(A) \to \mathbf{pStRing}(A)$, given by

$$(B, \alpha) \mapsto \left(\mathrm{End}^A_\bullet(B), \mathrm{End}^A_\bullet(\alpha)\right)$$

Note that there is a map $\eta : \mathrm{id} \to T$, given by $\eta(B, \alpha) = \alpha : B \to \mathrm{End}^A_\bullet(B)$ Then, we define

$$X^{(0)} = X, X^{(\alpha+1)} = T\big(X^{(\alpha)}\big), X^{(\lambda)} = \operatorname*{colim}_{\alpha<\lambda} X^{(\alpha)}, \lambda \text{ a limit cardinal}$$

We define $L(X) = X^{(\aleph_1)}$. Since T commutes with $\aleph_1$-filtered colimits, and the cardinal $\aleph_1$ is itself $\aleph_1$-filtered, we have

$$T(L(X)) = T\big(X^{(\aleph_1)}\big) = T\Big(\operatorname*{colim}_{\alpha<\aleph_1} X^{(\alpha)}\Big) = \operatorname*{colim}_{\alpha<\aleph_1} T\big(X^{(\alpha)}\big) = X^{(\aleph_1)} = L(X)$$

Hence, $L(X) \in \mathbf{StRing}(A)$.

On the other hand, for any $Y \in \mathbf{StRing}(A)$, it suffices to show that for any $X \in \mathbf{pStRing}(A)$, we have

$$\eta_X^* : \mathrm{Hom}(T(X), Y) \to \mathrm{Hom}(X, Y)$$

is an equivalence. This is because $T(Y) = Y$, hence we can construct an inverse via sending $f : X \to Y$ to $\eta_Y^{-1} \circ T(f)$. □

Corollary 3.2.1.12. Colimits in $\mathbf{StRing}(A)$ are computed via spectrifying the colimit in $\mathbf{pStRing}(A)$.

Construction 3.2.1.13. Let $A \to B$, $A \to C$ be maps of Stefanich rings. We define $B \otimes_A C$ to be the pushout of the two maps. By construction, this is the spectrification of

$$((B/A)_0 \otimes (C/A)_0, (B/A)_1 \otimes (C/A)_1, \cdots)$$

Or, if we regard it as a ring over C, it is the spectrification of

$$(g_0^*(B/A)_0, g_1^*(B/A)_1, \cdots)$$

Likewise to the category $\mathbf{Sp}$, there is also a shift functor in $\mathbf{StRing}$.

Construction 3.2.1.14. Consider the forgetful functor $\mathbf{CAlg}((n+1)\mathbf{Pr}) \to \mathbf{CAlg}(n\mathbf{Pr})$ right adjoint to the inclusion $\mathbf{CAlg}(n\mathbf{Pr}) \hookrightarrow \mathbf{CAlg}((n+1)\mathbf{Pr})$ by directly regarding an n-category as an $(n+1)$-category. This is the functor of taking cores. Thus, we have a map of shifting given by taking the colimit

$$\begin{array}{ccccc}
\cdots & \longrightarrow & \mathbf{CAlg}(2\mathbf{Pr}) & \longrightarrow & \mathbf{CAlg}(1\mathbf{Pr}) \\
\downarrow{\scriptstyle (-)^\simeq} & & \downarrow{\scriptstyle (-)^\simeq} & & \downarrow{\scriptstyle (-)^\simeq} \\
\cdots & \longrightarrow & \mathbf{CAlg}(1\mathbf{Pr}) & \longrightarrow & \mathbf{CAlg}(0\mathbf{Pr})
\end{array}$$

We denote this functor by $\vec{\Omega} : \mathbf{StRing} \to \mathbf{StRing}$. This functor is by definition fully faithful and has essential image given by $\mathbf{StRing}_{/\vec{\Omega}(\mathbb{P})}$, where

$$\vec{\Omega}(\mathbb{P}) = \vec{\Omega}(*, 0\mathbf{Pr}, 1\mathbf{Pr}, \cdots) = (0\mathbf{Pr}, 1\mathbf{Pr}, 2\mathbf{Pr}, \cdots)$$

3.2.2. Affine, proper, and étale maps

Definition 3.2.2.1. Let $f : A \to B$ be a map in **StRing**. We say it is n-affine if B is in the essential image of the inclusion $\mathbf{CAlg}(A_{n+1}) \hookrightarrow \mathbf{StRing}(A) = \mathbf{StRing}_{A/}$.

By definition, this means that for $m \geqslant n$, we have $(B/A)_{m+1} = \mathbf{Mod}^A_{m+1}((B/A)_m)$.

We first give some basic properties of n-affine maps.

Lemma 3.2.2.2. Let $B \in \mathbf{StRing}_{A/}$ be n-affine. Then, the comparison map $\mathbf{Mod}_{(B/A)_m}(A_{m+1}) \to B_{m+1}$ is an equivalence.

Proof: Take any $m \geqslant n$. Recall the pair of adjunction

$$L^A_{m+1} : \mathbf{Mod}_{A_{m+1}}((m+1)\mathbf{Pr}) \rightleftarrows A_{m+2} : U^A_{m+1}$$

By definition, we have

$$\begin{aligned}
B_{m+1} &= \mathrm{End}_{B_{m+2}}\left(1_{B_{m+2}}\right) \\
&= \mathrm{Hom}_{A_{m+2}}\left(1_{A_{m+2}}, f_{m+1,*}1_{B_{m+2}}\right) && \text{by adjunction} \\
&= U^A_{m+1} f_{m+1,*} 1_{B_{m+2}} && \text{by definition of } U^A_{m+1}, \text{ Construction 3.2.1.2} \\
&= U^A_{m+1}(B/A)_{m+1} && \text{by definition of } (B/A)_{m+1}, \text{ Construction 3.2.1.6} \\
&= U^A_{m+1}\mathbf{Mod}^A_{m+1}((B/A)_m) && \text{by affineness} \\
&= U^A_{m+1}L^A_{m+1}\mathbf{Mod}_{(B/A)_m}(A_{m+1})
\end{aligned}$$

Hence, the comparison map in question is actually the unit map of $L^A_{m+1} \dashv U^A_{m+1}$ on the object $\mathbf{Mod}_{(B/A)_m}(A_{m+1})$. By Lemma 3.2.1.4, it suffices to show that $\mathbf{Mod}_{(B/A)_m}(A_{m+1})$ is a $\aleph_1$-filtered colimit of dualizable objects, which is done by writing $(B/A)_m$ as an $\aleph_1$-filtered colimit of its countably presented subalgebras. □

Corollary 3.2.2.3. Let $g : A \to C$ be any map of Stefanich rings. Then, for any n-affine A-algebra $A \to B$, we have $g^*_m(B/A)_m = (B \otimes_A C/C)_m$ for $m \geqslant n$. Equivalently, the spectrification process in Construction 3.2.1.13 to define the tensor product $B \otimes_A C/C$ is not needed.

Proof: Note that by affineness, we have $g^*_{m+1} \circ \mathbf{Mod}^A_m = \mathbf{Mod}^C_m \circ g^*_m$ for $m \geqslant n$. Thus, presenting $g^*(B)$ as the spectrification of the pre-Stefanich ring

$$(g^*_0(B/A)_0, g^*_1(B/A)_1, \cdots)$$

by the above relation, this is already a Stefanich ring if we omit the part where $i \leqslant n$. Hence, $g^*(B)$ is an n-affine ring over C determined by $g^*_n(B/A)_n$. Hence, by a direct computation, we have $g^*_m(B/A)_m = (B \otimes_A C/C)_m$ for $m \geqslant n$. □

Corollary 3.2.2.4. Let $h = g \circ f : A \to B \to C$. If f is n-affine, then h is n-affine iff g is.

Proof: We first give a better characterization of the condition where h is n-affine. Notice that C is n-affine over A iff it lies in the essential image of $\mathbf{CAlg}(A_{n+1}) \hookrightarrow \mathbf{StRing}_{A/}$. Slicing over the object $(B/A)_n$, we obtain a functor of categories $\mathbf{CAlg}(A_{n+1})_{(B/A)_n/} \hookrightarrow \mathbf{StRing}_{B/}$. Thus, a ring $B \hookrightarrow C$ is n-affine over A iff it lies in the slice $\mathbf{CAlg}(A_{n+1})_{(B/A)_n/}$. However, by affineness, we may identify $B_{n+1} = \mathbf{Mod}_{(B/A)_n}(A_{n+1})$ by Lemma 3.2.2.2, so

$$\mathbf{CAlg}(A_{n+1})_{(B/A)_n/} = \mathbf{CAlg}(B_{n+1})$$

Thus, the essential image of h being n-affine and g being n-affine coincide. □

We then move on to proper and étale morphisms.

Definition 3.2.2.5. Let $f : A \to B$ be a map of Stefanich rings. We say it is n-prim if for all $m \geqslant n$, the map $(f/A)^*_m : (A/A)_{m+1} \to (B/A)_{m+1}$ admits a right adjoint $(f/A)_{m,*} : (B/A)_{m+1} \to (A/A)_{m+1}$ in the category A_{m+2}.

By definition, if we compose two n-prim maps, we can apply the forgetful functor $B_{m+1} \to A_{m+1}$ and compose the right adjoint, and conclude that the composition is still n-prim.

Lemma 3.2.2.6. The composition $(f/A)_{m,*}(f/A)^*_m : (A/A)_{m+1} \to (A/A)_{m+1} \in \mathrm{End}(A_{m+2})$ is identified with $(B/A)_m$.

Proof: Note that the functor $A_{m+2} \to (m+2)\mathbf{Pr}$ preserves adjunctions, so the adjunction $(f/A)^*_m \dashv (f/A)_{m,*}$ is send to the adjunction $f^*_m \dashv f_{m,*}$, and the latter sends $1_{A_{m+1}}$ to $(B/A)_{m+1}$ by definition. □

Proposition 3.2.2.7. If $f : A \to B$ is n-affine and countably presented, then it is n-prim.

Proof: The map $(f/A)^*_m : (A/A)_{m+1} \to (B/A)_{m+1}$ factors as

$$(A/A)_{m+1} \to \mathbf{Mod}^A_{m+1}((B/A)_m) \xrightarrow{L^{B/A}_m} (B/A)_{m+1}$$

by definition. Then, for $m \geqslant n$, since f is n-affine, $L^{B/A}_m$ is an equivalence. Hence, it suffices to show that the first functor admit a right adjoint.

Notice that this functor is the image of $A_{m+1} \to \mathbf{Mod}_{(B/A)_m}(A_{m+1})$ under the functor

$$L^A_{m+1} : \mathbf{Mod}_{A_{m+1}}((m+2)\mathbf{Pr}) \to A_{m+2}$$

The functor L^A_{m+1} preserves adjoints, and the above free functor $F_{(B/A)_m}$ admit a right adjoint in $\mathbf{Pr}^L_\kappa$ iff $A \to B$ is κ-finitely presented. In this case, since $A \to B$ is countably presented, there is a right adjoint of $A_{m+1} \to \mathbf{Mod}_{(B/A)_m}(A_{m+1})$ internal to $\mathbf{Mod}_{A_{m+1}}((m+2)\mathbf{Pr})$. Thus we obtain the desired right adjoint. □

Lemma 3.2.2.8. If $f : A \to B$ is n-prim, then for any $g : A \to C$, we have the induced pushout map $f' : C \to C \otimes_A B$ n-prim. Moreover, for every $m \geqslant n$, the comparison map

$$g_m^*(B/A)_m \to (B \otimes_A C/C)_m$$

is an isomorphism. Equivalently, the spectrification process in Construction 3.2.1.13 to define the tensor product $B \otimes_A C/C$ is not needed.

Proof:

□

Lemma 3.2.2.9. A filtered colimit of n-prim A-algebras is n-prim.

Proof: Let I be a filtered index category with an initial object, and let A_i □

3.2.3. The category of Gestalten

3.3. Six functor formalisms and Gestalten

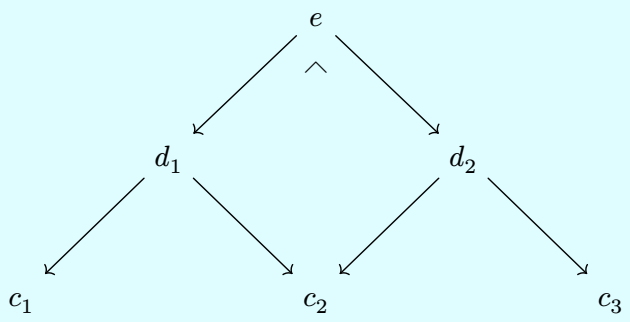
3.3.1. Six functor formalisms

Definition 3.3.1.1. We define a geometric setting to be a pair $(\mathcal{C}, E)$, where $\mathcal{C}$ is a category with finite limits and $E \subset \mathcal{C}$ is a wide subcategory (i.e. a class of morphisms) subject to the following conditions:
(1) E contains all isomorphisms.
(2) For $f : c \to d \in E$, for any $g : e \to d \in \mathcal{C}$ we have $g^* f : c \times_d e \to e \in E$.
(3) For $f : c \to d, g : d \to e \in E$, we have $g \circ f : c \to e \in E$.
(4) For $f : c \to d \in E$, we have $\Delta_f : c \to c \times_d c \in E$.

Construction 3.3.1.2. Let $(\mathcal{C}, E)$ be a geometric setup. We define $\mathrm{Corr}(\mathcal{C}, E)$ to be the following 2-category:
(1) The objects of $\mathrm{Corr}(\mathcal{C}, E)$ are the same ones as $\mathcal{C}$.
(2) For $c, d \in \mathcal{C}$, we define $\mathrm{Hom}_{\mathrm{Corr}(\mathcal{C},E)}(c, c')$ to be the category of diagrams $c \leftarrow d \to c'$ in $\mathcal{C}$, where the map $d \to c' \in E$.
(3) The composition of morphisms are given by pullbacks. Namely, for morphisms

$$f_1 = c_1 \leftarrow d_1 \to c_2, \quad f_2 = c_2 \leftarrow d_2 \to c_3$$

we define their composition to be the pullback $c_1 \leftarrow d_1 \times_{c_2} d_2 \to c_3$, to be depicted in the following diagram:



Here, $e \to d_2$ is in E by the axiom (2), and the composition $e \to c_3$ is in E by the axiom (3).

We notice that here we have $c \leftarrow d \to c'$ is the composition of $c \leftarrow d \to d$ and $d \leftarrow d \to c'$.

More rigorously, we can realize $\mathrm{Corr}(\mathcal{C}, E)$ as a 1-full subcategory of $\mathbf{Cat}_{/\mathcal{C}}$, where:
(1) $c \in \mathcal{C}$ is mapped to $\mathcal{C}_{/c}$.
(2)

In particular, if E is the set of all morphisms in $\mathcal{C}$, we denote $\mathrm{Corr}(\mathcal{C})$ by $\mathrm{Corr}(\mathcal{C}, E)$.

Construction 3.3.1.3. There is a symmetric monoidal structure on $\mathrm{Corr}(\mathcal{C}, E)$. For $c, d \in \mathrm{Corr}(\mathcal{C}, E)$, we define $c \otimes d = c \times d$, where the product is taken in $\mathcal{C}$. The product on morphisms is likewise defined.

Construction 3.3.1.4. We note that $\mathrm{Corr}(\mathcal{C})$ admits an enrichment by itself. This enrichment comes from the fact that the symmetric monoidal structure $\mathrm{Corr}(\mathcal{C})^{\times}$ is closed symmetric monoidal. We will denote the enriched version by $\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C})$ if we wish to emphasize that the category we are using is the enriched one.

For $X \to * \in E$, note that the morphisms $* \leftarrow X \to X \times X$ and $X \times X \leftarrow X \to *$ presents X as the dual of itself (directly). Thus, we may set the internal Hom of $\mathrm{Corr}(\mathcal{C})$ by $\underline{\mathrm{Hom}}(X, Y) = X \times Y$.

As is so for general closed symmetric monoidal categories, we have

$$\mathrm{Hom}(X, Y) = \mathrm{Hom}(\mathbb{1}, \underline{\mathrm{Hom}}(X, Y))$$

In this setting, unwinding the definitions, the right hand side are the space of spans

$$* \leftarrow Z \to X \times Y$$

and $Z \to X \times Y$ is equivalent to the data of $X \leftarrow Z \to Y$, i.e., a morphism $X \to Y$ in $\mathrm{Corr}(\mathcal{C})$.

Definition 3.3.1.5. Let $(\mathcal{C}, E)$ be a geometric setup. A 3-functor formalism is a lax symmetric monoidal functor

$$D : \mathrm{Corr}(\mathcal{C}, E) \to \mathbf{Cat}$$

A 6-functor formalism is a lax symmetric monoidal functor

$$D : \mathrm{Corr}(\mathcal{C}, E) \to \mathbf{Pr}^{L}$$

We will denote the image of $X \in \mathcal{C}$ by $D(X)$. For any morphism $f : X \to Y \in \mathcal{C}$, we define the image of $Y \leftarrow X \to X$ by D as $f^* : D(Y) \to D(X)$, and for any morphism $g : X \to Y \in E$, we define the image of $X \leftarrow X \to Y$ by D as $g_! : D(X) \to D(Y)$.

Definition 3.3.1.6. We then define the 2-category of geometric settings. Given geometric settings $(\mathcal{C}, E)$ and $(\mathcal{C}', E')$, we define the category of morphisms $\mathrm{Hom}_{\mathrm{GeomSetup}}((\mathcal{C}, E), (\mathcal{C}', E'))$ to be the 1-full subcategory of $\mathrm{Fun}(\mathcal{C}, \mathcal{D})$, where a functor $F : \mathcal{C} \to \mathcal{D}$ is a morphism of geometric settings iff the following conditions holds:

(1) $\varphi(E) \subset F$.
(2) φ preserves fibered products $c \times_d e$ where $c \to d \in E$.

Here, the intuition of the second condition is that φ preserves the composition in $\mathrm{Corr}(\mathcal{C}, E)$.

Then, a natural transformation $\omega : F \to F'$ of morphisms of geometric settings lies in $\mathrm{Hom}_{\mathrm{GeomSetup}}((\mathcal{C}, E), (\mathcal{C}', E'))$ if it

todo 3.3.1.1. Nagata settings.

3.3.2. The 2-category of kernels and higher categorical sheaves

In this section, we will use many facts on enriched categories. We will not record the exact details of how to define enriched categories in the ∞-categorical setting, and we refer the readers to [HM24].

Definition 3.3.2.1. Let $\mathcal{V}$ be a monoidal category. We define $\mathbf{Cat}^{\mathcal{V}}$ to be the $(\infty, 1)$-category of categories enriched in $\mathcal{V}$.

Construction 3.3.2.2. Let $A \in \mathbf{Alg}(\mathcal{V})$. Then, there is an object $BA \in \mathbf{Cat}^{\mathcal{V}}$, with one object $*$ and $\mathrm{End}(*) = A$. This can be regarded as a functor $B : \mathbf{Alg}(\mathcal{V}) \to \mathbf{Cat}^{\mathcal{V}}_*$, the target being the category of pointed $\mathcal{V}$-enriched categories. This functor admits a right adjoint, given by

$$\Omega : \mathbf{Cat}^{\mathcal{V}}_* \to \mathbf{Alg}(\mathcal{V}), (\mathcal{C}, c) \mapsto \mathrm{End}_{\mathcal{C}}(c)$$

We first state the main categorical tool we will use in this section (which we will not prove).

Theorem 3.3.2.3. Let $\varphi : \mathcal{V} \to \mathcal{W}$ be a lax monoidal morphism of monoidal categories. Then, there exists a transfer functor

$$\tau_{\varphi} : \mathbf{Cat}^{\mathcal{V}} \to \mathbf{Cat}^{\mathcal{W}}$$

having the intuition of sending a category $\mathcal{C}$ to a category enriched in $\mathcal{W}$ with the same objects and

$$\mathrm{Hom}^{\mathcal{W}}_{\mathcal{C}}(c, d) = \varphi\big(\mathrm{Hom}^{\mathcal{V}}_{\mathcal{C}}(c, d)\big)$$

Definition 3.3.2.4. Given a 3-functor formalism $D : \mathrm{Corr}(\mathcal{C}, E) \to \mathbf{Cat}$, we can define the category of kernels as follows. Firstly, if we have E the set of all morphisms in $\mathcal{C}$, we define

$$\mathcal{K}_D := \tau_D(\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C})) \in \mathbf{Cat}^{\mathbf{Cat}} = \mathbf{Cat}_2$$

where $\mathrm{Corr}(\mathcal{C}) \in \mathbf{Cat}^{\mathrm{Corr}(C)}$ via the closed symmetric monoidal structure. Then, in the general case, for $S \in \mathcal{C}$, we define

$$\mathcal{K}_{D,S} = \tau_D\Big(\mathrm{Corr}^{\mathrm{enr}}\Big((\mathcal{C}_E)_{/S}\Big)\Big) \in \mathbf{Cat}_2$$

Explicitly, $\mathcal{K}_{D,S}$ has objects $X \to S \in E$, and $\mathrm{Hom}(X \to S, Y \to S) = D(X \times_S Y)$. The composition of morphisms are given by convolution. Namely, for $X, Y, Z \in (\mathcal{C}_E)_{/S}$, and for morphisms $X \to Y$ presented by $\mathcal{F} \in D(X \times_S Y)$ and $Y \to Z$ presented by $\mathcal{G} \in D(Y \times_S Z)$, we define their composition to be

$$\pi_{X,Z,!}\left(\pi^*_{X,Y}\mathcal{F} \otimes \pi^*_{Y,Z}\mathcal{G}\right)$$

remark 3.3.2.1. The seemingly correct way of understanding this definition is via Fourier-Mukai transforms. Recall that any object $\mathcal{K} \in D(X \times_S Y)$ gives rise to a functor $D(X) \to D(Y)$ via $\mathcal{F} \mapsto \pi_{Y,!}(\mathcal{K} \otimes \pi^*_X \mathcal{F})$, where $\mathcal{K}$ acts as something like the convolution kernel.

We state another theorem of kernel categories that we will not prove, since we will prove a more general version directly.

Theorem 3.3.2.5. Let D be a 3-functor formalism on a geometric setup $(\mathcal{C}, E)$. Then, the assignment $S \mapsto \mathcal{K}_{D,S}$ upgrades to a lax symmetric monoidal functor

$$\mathcal{K}_{D,(-)} : \mathrm{Corr}(\mathcal{C}, E) \to \mathbf{Cat}_2$$

Moreover, there exists a natural transformation

$$\Phi_{D,(-)} : \mathrm{Corr}\left((\mathcal{C}_E)_{/(-)}\right) \to \mathcal{K}_{D,(-)}$$

Taken at a value S, we have $\Phi_{D,(S)}$ sends an object $X \to S$ to $X \to S$, and sends

$$X \overset{f}{\leftarrow} Z \overset{g}{\to} Y \mapsto (f,g)_!\left(1_{D(Z)}\right) \in D(X \times_S Y)$$

Then, Stefanich generalized the above definitions to a version of "n-categories of kernels" in the paper [Ste25].

Definition 3.3.2.6. Now, suppose $\mathcal{V}$ is an symmetric monoidal category. We define $\mathbf{Cat}^{\mathcal{V}}_n$ inductively as follows:
(1) For $n = 1$, $\mathbf{Cat}^{\mathcal{V}}_n \coloneqq \mathbf{Cat}^{\mathcal{V}}$.
(2) For $n > 1$, $\mathbf{Cat}^{\mathcal{V}}_n \coloneqq \mathbf{Cat}^{\mathbf{Cat}^{\mathcal{V}}_{n-1}}$

Here we use the fact that for $\mathcal{V}$ a symmetric monoidal category, $\mathbf{Cat}^{\mathcal{V}}$ admits a symmetric monoidal structure. Heuristically, this symmetric monoidal structure sends $\mathcal{C}, \mathcal{D} \in \mathbf{Cat}^{\mathcal{V}}$ to a category with objects pairs (c, d) for $c \in \mathcal{C}, d \in \mathcal{D}$, and

$$\mathrm{Hom}^{\mathcal{V}}((c_1, d_1), c_2, d_2) = \mathrm{Hom}^{\mathcal{V}}(c_1, c_2) \otimes \mathrm{Hom}^{\mathcal{V}}(d_1, d_2)$$

In particular, the monoidal unit is the category with one object $*$, and $\mathrm{End}(*) = 1_{\mathcal{V}}$. This is the category $B1_{\mathcal{V}}$.

We will use a further fact on enriched categories.

Theorem 3.3.2.7. Let $\mathcal{M}$ be a presentably symmetric monoidal category, and let $A \in \mathbf{CAlg}(\mathcal{M})$. Then, we have a symmetric monoidal equivalence

$$\Gamma_A : \mathbf{Cat}^{\mathbf{Mod}_A(\mathcal{M})} \to \mathbf{Mod}_{BA}\big(\mathbf{Cat}^{\mathcal{M}}\big)$$

where Γ_A is constructed via the inclusion $\iota : \mathbf{Mod}_A(\mathcal{M}) \hookrightarrow \mathcal{M}$, namely

$$\tau_\iota : \mathbf{Cat}^{\mathbf{Mod}_A(\mathcal{M})} \to \mathbf{Cat}^{\mathcal{M}}$$

lies in the category of $\tau_\iota(1_{\mathbf{Cat}^{\mathbf{Mod}_A(\mathcal{M})}}) = BA$-modules.

Corollary 3.3.2.8. The iterated functor

$$\Gamma_A^n : \mathbf{Cat}_n^{\mathbf{Mod}_A(\mathcal{M})} \to \mathbf{Mod}_{B^n A}\big(\mathbf{Cat}_n^{\mathcal{M}}\big)$$

is an equivalence.

Construction 3.3.2.9. We may iteratively generalize the definition of $n\mathrm{Corr}(\mathcal{C})$. For $n = 1$, we define it simply to be $\mathrm{Corr}(\mathcal{C})$. For $n > 1$, we define it to be the category the same as $\mathcal{C}$, and such that

$$\mathrm{Hom}_{n\mathrm{Corr}(\mathcal{C})}(X, Y) = (n-1)\mathrm{Corr}\big(\mathcal{C}_{\mathcal{X}/-/\mathcal{Y}}\big)$$

It is a monoidal category via $(X, Y) \mapsto X \times Y$.

Construction 3.3.2.10. Via the adjunction $B \dashv \Omega$, we obtain the iterated counit map

$$B^{n-1}\mathrm{Corr}(\mathcal{C}) = B^{n-1}\Omega^{n-1}n\mathrm{Corr}(\mathcal{C}) \to n\mathrm{Corr}(\mathcal{C})$$

Which gives an enrichment $n\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C}) \in \mathbf{Cat}_n^{\mathrm{Corr}(\mathcal{C})}$.

Construction 3.3.2.11. Now, given a six-functor theory $\mathrm{Sh} : \mathrm{Corr}(\mathcal{C}) \to 1\mathbf{Pr}$ (i.e. Sh is a lax monoidal functor), we may extend it to an n-categorical sheaf via the following process.

Firstly, we define the target of the n-categorical sheaf theory. Note that there is a free (cocompletion) functor

$$\mathrm{cplt}_n : \mathbf{Cat}_n^{\mathbf{Pr}} \to (n+1)\mathbf{Pr}$$

Note that intuitively $\mathbf{Cat}_n^{\mathbf{Pr}}$ has the objects certain $(n+1)$-categories, since we have the inclusion $\mathbf{Cat}_n^{\mathbf{Pr}} \to \mathbf{Cat}_n^{\mathbf{Cat}} = \mathbf{Cat}_{n+1}$. Then, we define $n\mathbf{Pr}_{\mathrm{Sh}} \in (n+1)\mathbf{Pr}$ to be the category

$$n\mathbf{Pr}_{\mathrm{Sh}} := \mathrm{cplt}_n(\tau_{\mathrm{Sh}}(n\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C})))$$

Then, for the whole sheaf theory, notice that there is a natural transformation

$$\eta : U_{\mathrm{Corr}(\mathcal{C})} \to \mathrm{Sh} \in \mathrm{Fun}(\mathrm{Corr}(\mathcal{C}), \mathbf{Cat})$$

where $U_{\mathrm{Corr}(\mathcal{C})} : \mathrm{Corr}(\mathcal{C}) \to \mathbf{Ani}$ is given by $X \mapsto \mathrm{Hom}(\mathbb{1}, X)$. On the level of objects, this natural transformation is given by $(f : Z \to X) \mapsto f_*\left(\mathbb{1}_{\mathrm{Sh}(Z)}\right)$.

Hence, we have an induced functor

$$n\mathrm{Sh} : n\mathrm{Corr}(\mathcal{C}) = \tau_{U_{\mathrm{Corr}(\mathcal{C})}}(n\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C})) \to \tau_{\mathrm{Sh}}(n\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C})) \to \mathrm{cplt}_n(\tau_{\mathrm{Sh}}(n\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C}))) = n\mathbf{Pr}_{\mathrm{Sh}}$$

which is our desired higher analogue of the construction of the 2-category of kernels.

By construction, we may identify

$$(n-1)\mathrm{Sh} = \mathrm{End}(n\mathrm{Sh})$$

where $\mathrm{End}(\mathcal{C}) = \mathrm{End}_{\mathcal{C}}(\mathbb{1}_{\mathcal{C}})$, as in the previous definitions. Thus, this construction gives rise from a sheaf theory $\mathrm{Sh} : \mathrm{Corr}(\mathcal{C}) \to \mathbf{Pr}$ to a map of Gestalten

$$(1\mathrm{Sh}, 2\mathrm{Sh}, \cdots) : (1\mathrm{Corr}(\mathcal{C}), 2\mathrm{Corr}(\mathcal{C}), \cdots) \to (1\mathbf{Pr}_{\mathrm{Sh}}, 2\mathbf{Pr}_{\mathrm{Sh}}, \cdots)$$

However, in [Sch25], Scholze used a different iterative construction. This construction is levelwise easier, however loses information on the functoriality (for example, his sheaf theory $n\mathrm{Sh}$ still has source Corr instead of $n\mathrm{Corr}$). We record his construction as follows, and give an equivalence of the category of sheaves.

remark 3.3.2.2. We may recover the original sheaf theory from the map of Gestalten. We only need to consider the map $1\mathrm{Sh} : 1\mathrm{Corr}(\mathcal{C}) \to 1\mathbf{Pr}_{\mathrm{Sh}}$. By construction, we have $1\mathrm{Corr}(\mathcal{C}) = \mathrm{Corr}(\mathcal{C})$, and

$$1\mathbf{Pr}_{\mathrm{Sh}} = \mathrm{cplt}_1(\tau_{\mathrm{Sh}}(\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C})))$$

Unwinding the definitions, $\tau_{\mathrm{Sh}}(\mathrm{Corr}^{\mathrm{enr}}(\mathcal{C}))$ is the category of kernels of the 3-functor formalism $\mathrm{Sh} : \mathrm{Corr}(\mathcal{C}) \to \mathbf{Cat}$ (we forget the structure of presentability). Hence by construction, we have for $X \in \mathcal{C}$, $\mathrm{Sh}(X) = \mathrm{Hom}_{1\mathbf{Pr}_{\mathrm{Sh}}}\left(\mathbb{1}_{1\mathbf{Pr}_{\mathrm{Sh}}}, 1\mathrm{Sh}(-)\right)$.

4. Appendix

4.1. Inputs from homotopy theory

4.1.1. Phantom maps

The main purpose of this section is to prove the following theorem.

Theorem 4.1.1.1. Any composition of two phantom maps is zero.

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